English

(1001CJA101021230071)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 195

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. Free ^{238}U nuclei kept in a train emit alpha particles. When the train is stationery and a uranium nucleus decays, a passenger measures that the separation between the alpha particle and the recoiling nucleus becomes x in time t after the decay. If a decay takes place when the train is moving at a uniform speed v, the distance between the alpha particle and the recoiling nucleus at a time t after the decay, as measured by the passenger will be :-

(A) x + vt

(B) x – vt

(C) x

(D) depends on the direction of the train.

2. Two identical spherical masses are kept at some distance as shown. Potential energy when a mass m is taken from surface of one sphere to the other.



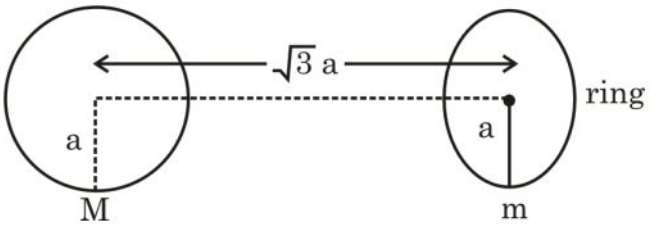
(A) Increases continuously.

(B) Decreases continuously.

(C) First increase then decreases.

(D) First decreases then increases.

3. A uniform ring of mass m and radius a and a sphere of mass M and radius a are released in free space. Consider only gravitational force between sphere and ring. Find acceleration of ring at given separation.



(A) $\frac{\sqrt{3}GM}{8a^3}$

(B) $\frac{\sqrt{3}GM}{4a^2}$

(C) $\frac{\sqrt{3}GM}{8a^2}$

(D) $\frac{GM}{\sqrt{3}a^2}$

4. A car drives along a road that has small depressions regularly spaced about 8.0m apart. When four 80 kg passengers enter the car, it sinks down by 1.6 cm. At approximately what speed the car might travel so that travelling in the vehicle becomes very uncomfortable?

(A) 5π m/s

(B) 10π m/s

(C) 2π m/s

(D) 20π m/s

5. Three tuning forks in the audible range produce beats: B and C produce beats at 7 Hz, A and C produce beats at 8 Hz. The frequency of B is 6% higher than that of A.

(A) Frequency of tuning fork A is 265 Hz.

(B) Frequency of tuning fork B is 265 Hz.

(C) Frequency of tuning fork C is 250 Hz.

(D) None of the above

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

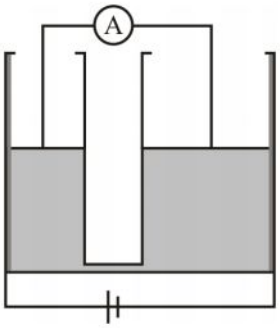
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

6. Two identical thin homogeneous nichrome wires are mounted into the side non-conducting walls of vertical communicating vessels (vessel cross-sectional area S = 100 cm^2 and 3S). The length of the wires is equal to the height of the vessels. An ideal battery with a voltage at the terminals U = 46.2 V is connected to the lower ends of the wires. The vessels are filled upto a level 50 cm and covered with light and thin conductive pistons. The pistons do not allow water to pass through, have contact with the wires and can move without friction inside the vessels without leaving them, thanks to stoppers.

An ammeter is connected to the pistons using flexible wires (the length of the wires allows the pistons to move freely). The installation diagram is shown in the figure. When a non-conducting weight weighing 1 kg is placed on a small piston, the ammeter shows a current strength of ℓ_1 = 2.31 A. If 3 such weights are placed on a large piston, the ammeter shows a current strength of ℓ_2 = 2.1 A.

We can assume that the connecting tube has a negligibly small cross-section. The conductivity of water is also very low. Water density ρ = 1000 kg/m^3.



(A) The resistance of the ammeter R_a is 1 ohm.

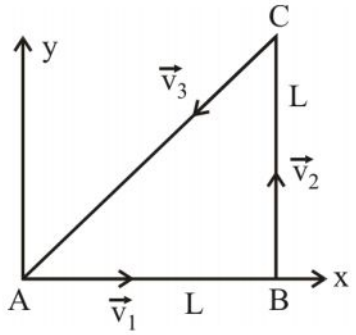
(B) The reading of ammeter when no weight is placed on the pistons is approx. 2.2 A.

(C) The maximum current through the ammeter is approx. 3.22 A.

(D) The minimum current through the ammeter is approx. 1.5 A.

7. Three tigers simultaneously begin to move along a horizontal surface with constant speeds. The velocity of the first tiger at any moment of time is directed towards the second, the velocity of the second towards the third, and the velocity of the third towards the first. At the initial moment of time, tigers form a right triangle with legs equal to L (Figure). Consider the size of the tigers to be much smaller than L. The speed of the first tiger is v, and the speeds of the second and third tigers v_2 and v_3 are such that during the movement the angles in the triangle ABC formed by the tigers remain constant.

Let's introduce a coordinate system as shown in the figure. The origin of coordinates coincides with the position of the first tiger at the moment of start (point A).



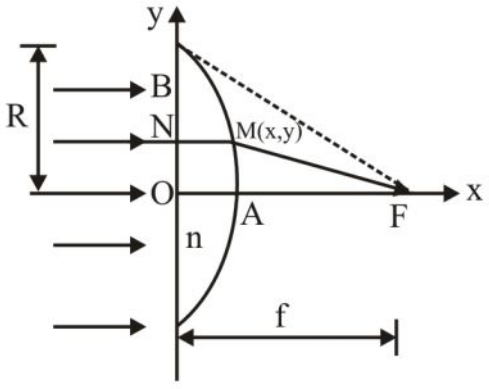
(A) The time after which the tigers will meet is $\frac{L}{v}$.

(B) The speed of second tiger v_2 is $\frac{v}{2}$.

(C) The speed of third tiger v_3 is $\frac{v}{2}$.

(D) The coordinates of the point where the tigers will meet is $\left(\frac{4L}{5}, \frac{L}{5}\right)$

8. As shown in figure, a plano-convex lens has a refractive index n and is placed in the air. The radius of the lens aperture is R. Take a point F on the principal optical axis of the lens, OF = f. When parallel light is incident along the principal optical axis, all the light rays converge at point F.



(A) The shape of the convex surface of the lens should be hyperbola.

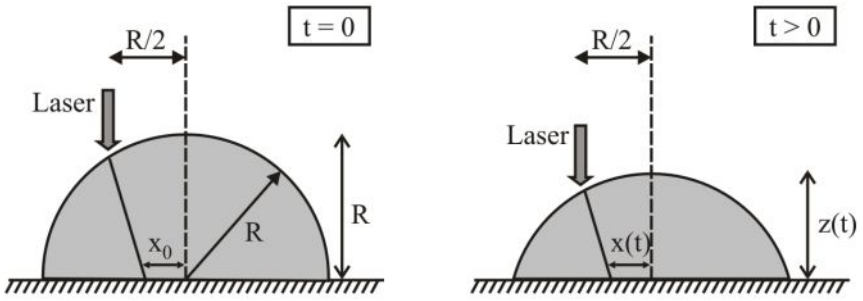
(B) The shape of the convex surface of the lens should be parabola.

(C) The maximum value of aperture R of the lens is nf.

(D) The maximum value of aperture R of the lens is $\sqrt{n^2-1}\,f$.

9. An ice hemisphere with radius R and refractive index n lies on a warm flat table and melts slowly. The rate of heat transfer from the table to the ice is proportional to the area of contact between them. It is known that the ice hemisphere completely melts in time T_0. Throughout the process, a laser beam is incident on the ice from above. The beam is vertically incident at a distance of R/2 from the axis of symmetry (see figure).

Assume that the temperature of the ice and the surrounding atmosphere are 0°C and remains constant during the melting process. The laser beam does not transfer energy to the ice. The melting water immediately flows off the table, and the ice does not move along the table.



(A) The value of x_0 (point on the table where the laser beam strikes at t = 0) is $\frac{2R}{\left(\sqrt{12n^2-3}+1\right)}$.

(B) The value of x_0 (point on the table where the laser beam strikes at t = 0) is $\frac{2R}{\left(\sqrt{4n^2-3}+1\right)}$.

(C) The height of ice z at any time t is $z = R\left(1-\frac{t^2}{T_0^2}\right)$.

(D) The time at which the value of x (point on the table where the laser beam strikes) becomes $\frac{R}{2}$ is $\frac{\sqrt{3}}{2}T_0$.

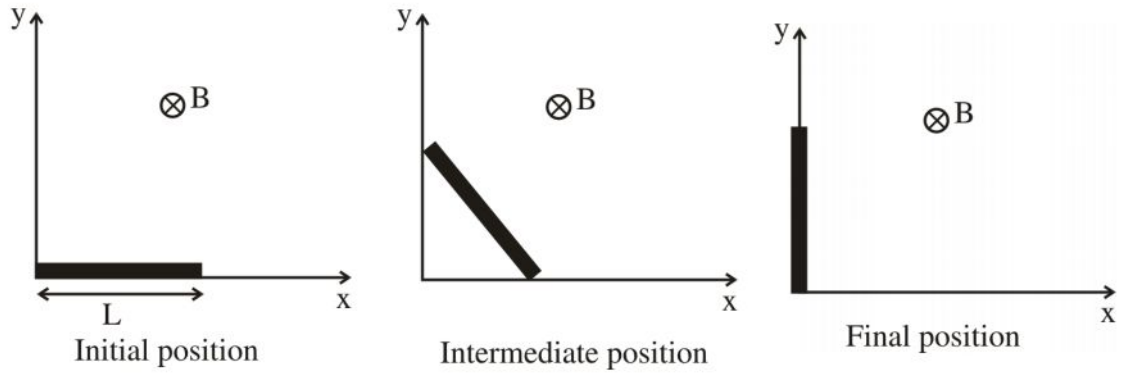
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.
Zero Marks : 0 In all other cases

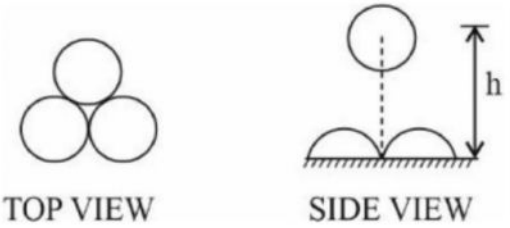
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1. A metallic rod of mass m and length L can slide without friction on two perpendicular wires. Entire arrangement is located in the horizontal plane. A magnetic field of magnitude B exists perpendicular to this plane in the downward direction. The wire have negligible resistance compared to the rod whose resistance is R. Initially rod is along one of the wire so that one end of it is at the junction of two wires show the rod is given an initial angular speed ω such that it slide with two ends in contact with the two wires & just comes to rest in an aligned position with the other wire. The value of ω is $\frac{3\pi B^2L^2}{\alpha mR}$. The value of α is :- (Neglect self inductance of system.)

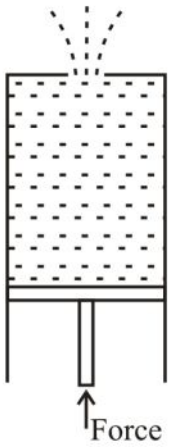


2. Two blocks of metal with heat capacities $C_1 = 3J/K$ and $C_2 = 6$ J/k, are interconnected by a rod of length ℓ and cross-sectional area S and fairly low heat conductivity K. The whole system is thermally insulated from the environment. At a moment t = 0 the temperature difference between the two blocks of metal equals $(\Delta T)_0$. Assuming the heat capacity of the rod to be negligible, if $\frac{\Delta T}{\Delta T_0} = e^{-\frac{SKt}{\alpha\ell}}$, then α is

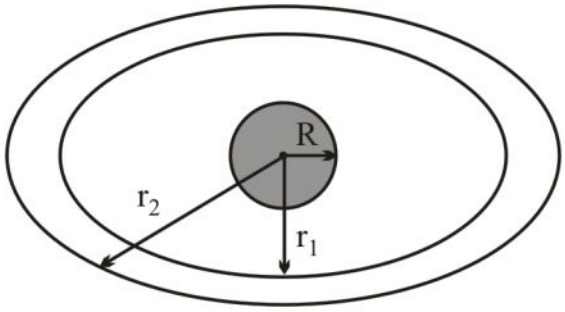
3. A sphere of mass m falls symmetrically on a set of three hemispheres placed in contact with each other on smooth horizontal ground from a height h. Mass of each hemisphere is m/2 and radius of each body is R. Neglect any rebounding of hemisphere. If sphere comes to rest after the simultaneous collision with three hemispheres then ratio of KE of sphere before collision and KE of each hemisphere after collision is p. Find the value of p.



4. A syringe is filled with water up to volume 20 cm^3. The area of cross-section of the cylinder is 5 cm^2. The syringe is held fixed vertically and its 90 gm piston is pushed upward by external agent with constant speed. A water beam coming out of the small nozzle (hole area 1 mm^2) has speed 2 m/s. Neglecting friction and viscous nature of water, find the work done by the agent (in 10^{-2} J) in fully emptying the syringe. (Take : g = 10 m/s^2)



5. A positive charge with a constant surface density σ is distributed over the surface of a thin flat non-conducting ring with internal and external radii of r_1 and r_2, respectively. The center of a non-conducting ball with radius $r < r_1$ coincides with the center of the ring. The sphere is uniformly charged in volume with a positive charge Q. At what values of σ (in C/m^2) is the resulting field strength vector at points on the surface of the ball located in the plane of the ring is zero? (Take : R = 1cm, r_1 = 2m, r_2 = 3m and $Q = \pi \times 10^{-6}$ C)



6. We use potential energy as mgh which is an approximate value, the actual value is $mgR^2\left(\frac{1}{R}-\frac{1}{R+h}\right)$. Where R is radius of earth (6400 km) and h is height from earth's surface. Find maximum height (in km) from earth's surface where the potential energy expression has error of less than 5% of true value.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

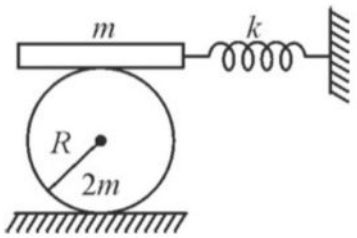
- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks : 0 If none of the options is chosen
Negative Marks : –1 In all other cases

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1. A uniform plank of mass m, free to move in the horizontal direction, is placed on the top of a solid cylinder of mass 2m and radius R. The plank is attached to a fixed wall with the help of a light spring of force constant k. The plank is slightly displaced towards spring and released. There is no slipping anywhere. For any displacement x, amplitude x_0 and velocity v of the plank, match the column-I with column-II and mark the correct option from the codes given below.



	Column-I		Column-II
(A)	Velocity of centre of cylinder.	(P)	$\frac{1}{2}kx_0^2$
(B)	Energy of oscillations of the system.	(Q)	$\frac{v}{2}$
(C)	Magnitude of acceleration of the plank.	(R)	$\frac{7}{8}mv^2+\frac{1}{2}kx^2$
(D)	Angular frequency of oscillations.	(S)	$\frac{4kx}{7m}$
		(T)	$\sqrt{\frac{4k}{7m}}$

2.

Column –I (Situation)		Column –II (Values in MKS units)	
(A)	Magnitude of electric field at (1,1,1) if potential is given by $V = -\frac{1}{x} - \frac{1}{y} - \frac{1}{z}$, is	(P)	0.4
(B)	The charge flowing through a resistance 6Ω varies with t as $Q = 4t - 2t^2$. The total heat produced in resistance is	(Q)	$\sqrt{3}$
(C)	A milliammeter of range 10mA has a coil of resistance 0.99 Ω. To use it as an ammeter of range 1A, the required shunt must have a resistance of	(R)	0.01
(D)	Two unlike point charges of same magnitude 10 mC are placed on x-axis at a separation of 8 cm. If both charges placed symmetrically about origin in external electric field $\vec{E} = 500V\,m^{-1}\,\hat{i}$, then potential energy of the system may be [Assume potential energy to be zero when dipole is perpendicular to electric field]	(S)	32
		(T)	None of these

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. Which of the following pair of complex both show stereo isomerism as well as structural isomerism.

(A) $[Pt(en)(CN)_2]$, $[Pt(en)_3]Cl_3$

(B) $[Co(H_2O)_3F_3]$, $[Co(NH_3)_4Cl_2]Cl$

(C) $[Pt(NH_3)_2Cl_2Br_2]$, $[Pt(NH_3)_5Cl)]Br$

(D) $[Co(H_2O)_4Cl_2]Cl.2H_2O$, $[Co(NH_2CH_2CH_2NH_2)_2Cl_2]NO_3$

2. Which of the following is not disproportionation reaction.

(A) $P_4 \xrightarrow{NaOH}$

(B) $HNO_2 \xrightarrow{\Delta}$

(C) $XeF_6 \xrightarrow{H_2O/OH^-\text{(excess)}}$

(D) $NaOCl \xrightarrow[\Delta]{\text{dil. HCl}}$

3. The rate of effusion of two gases 'A' and 'B' under identical conditions of temperature and pressure are in the ratio of 2 : 1. What is the ratio of rms velocity of their molecules if temperature of A & B are in ratio 2 : 1

(A) $\sqrt{2}$: 1

(B) 2 : 1

(C) 1 : $\sqrt{2}$

(D) $2\sqrt{2}$: 1

4.

A(Hydrocarbon) (C ⇒ 88.24%) [Molecular weight of A = 68]

$$A \xrightarrow[\text{n - propyl bromide}]{Na} B\,(C_8H_{14})$$

$$B \xrightarrow[\text{dil.}H_2SO_4]{Hg^{2+}} C_5H_6O\,(C)$$

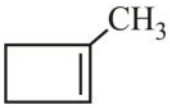
$$B \xrightarrow[H^+]{KMnO_4} \text{2-isomeric acids D and E.}$$

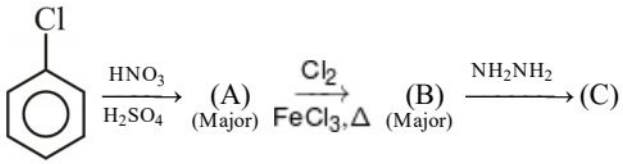
'A' is :

(A) $H_3C-H_2C-C\equiv C-CH_3$

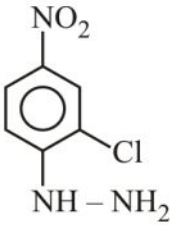
(B) $CH_3-CH_2-CH_2-C\equiv C-H$

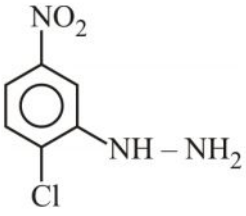
(C) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-C\equiv CH$

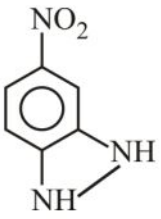
(D) 

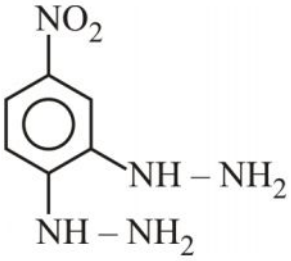
5. 

(C) is

(A) 

(B) 

(C) 

(D) 

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

6. Which of the following order is/are correct?

(A) RbCl > LiCl (Ionic character)

(B) $AlCl_3 > BCl_3$ (Lewis acidic strength)

(C) Atomic nitrogen > molecular nitrogen (Ionization energy)

(D) $Li_2CO_3 > Cs_2CO_3$ (Thermal stability)

7. In the homolytic cleavage of N_2O_4 which of the following propertie(s) changes.

(A) Polarity

(B) Bond length of NO

(C) Bond angle ($O - \hat{N} - O$)

(D) Magnetic behaviour

8. Which of the following is/are correct at 298K :

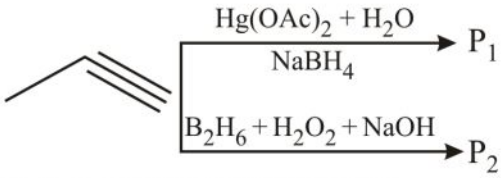
(A) $\Delta_f G°\ (H_{2(g)}) = 0$

(B) $\Delta_f G°\ (H_2O,\ \ell) = 0$

(C) $\Delta_f G°\ (D_2,\ g) = 0$

(D) $\Delta_f G°\ (H^+,\ aq.) = 0$

9.



P_1 & P_2 will be distinguished by

(A) 2, 4, D.N.P

(B) alkaline Cu^{2+} (tartarate)

(C) $NaHSO_3$

(D) Schiff's reagent

SECTION-II : (Maximum Marks: 18)

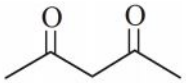
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.
Zero Marks : 0 In all other cases

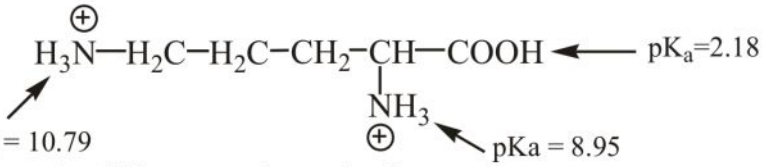
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1. Find the sum of element which does not have f-subshell e^-.
Th , La , Hg , Pb , Os , Pt , Rh

2. When CrI_3 reacts with H_2O_2 in presence of NaOH forming Na_2CrO_4, $NaIO_4$ & H_2O then find how many times will be rate of disappearance of NaOH compared to CrI_3.

3.  Number of iodoform molecules produced per molecule.

4. If pKa values for protonated Lysine molecule are $pKa_1 = 2.18$; $pKa_2 = 8.95$; $pKa_3 = 10.79$ respectively then , Calculate vlue of $(x - 1)$ where x is iso-electric point of lysine.



Answer should be nearest integral value

5. Identify reactions in which new C—C bond is formed ?

(i) Aldol (ii) Cannizaro (iii) Reimer Tiemann
(iv) Hoffmann's Bromamide (v) Sandmeyer reaction
(vi) Gattermann Aldehyde sysnthesis (vii) Friedel craft reaction
(viii) Wurtz reaction

6. Find the maximum number of electron(s) in an orbital which having n = 4 and $\ell = 2$.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

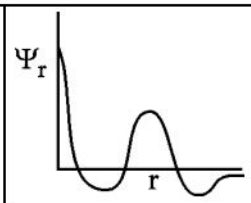
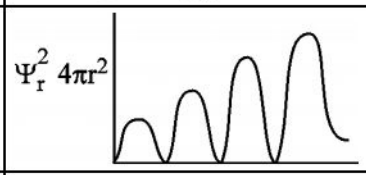
Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks : 0 If none of the options is chosen
Negative Marks : –1 In all other cases

---

1. Match the column :

Column-I		Column-II	
(A)	Tetragonal and Hexagonal	(P)	are two crystal systems
(B)	Cubic and Rhombohedral	(Q)	$a = b \neq c$
(C)	Monoclinic and Triclinic	(R)	$a \neq b \neq c$
(D)	Cubic and Hexagonal	(S)	$a = b = c$
		(T)	$\alpha = \beta = \gamma$

2. Column-I and Column-II contain data on Schrondinger Wave – Mechanical model, where symbols have their usual meaning. Match the columns :

Column-I		Column-II	
(A)		(P)	4s
(B)		(Q)	$5p_x$
(C)	$\Psi(\theta, \phi) = K$ (independent of θ & ϕ)	(R)	3s
(D)	Atleast one angular node is present	(S)	$6d_{xy}$
		(T)	$2p_z$

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. The value of $\lim\limits_{n\to\infty}\sum\limits_{n=1}^{n}\dfrac{n^2\tan^{-1}\left(\frac{2n+1}{1+(n^2+n)^2}\right)+(2n+1)\tan^{-1}(n+1)^2}{(n^2+n)^2\tan^{-1}n^2.\tan^{-1}(n+1)^2}$ is equal to -

(A) $-\dfrac{\pi}{4}$

(B) $\dfrac{4}{\pi}$

(C) $\dfrac{\pi}{2}$

(D) $\dfrac{2}{\pi}$

2. Let $A=[a_{ij}]_3$ be an orthogonal matrix, B and C are two other matrices of same order as A of which B is symmetric and C is skew symmetric such that $A(B+C)=KI$ where K is non zero scalar. If $Tr.(B)=8$ $Tr.(A)\neq 0$ then K is -
[Note : Tr.(P) denotes trace of matrix P]

(A) 2

(B) 3

(C) 4

(D) 8

3. If one number is selected from each of the three sets A = {1,2,3,4,5,6,7}, B = {1,2,3,4} and C = {2,3,4,5,6,7} then probability that product of the number selected from one and sum of the numbers selected from the other two is a prime, is -

(A) $\dfrac{1}{7}$

(B) $\dfrac{4}{7}$

(C) $\dfrac{13}{56}$

(D) $\dfrac{1}{14}$

4. Let $\vec{v}_1 = a^2\hat{i} - 2\hat{j} + (a+1)\hat{k}$ and $\vec{v}_2 = 2\hat{i} + \hat{j} - (a^2-1)\hat{k}$ be two vectors such that $\vec{v}_2$ lies in the plane $3x - \lambda y + z = 1$ and $(\vec{v}_1 + \vec{v}_2).\vec{v}_1 = a^4 + a^2 + 2a + 5$ then greatest volume of the parallelopiped whose co-terminous edges are represented by $\vec{v}_1, \vec{v}_2$ and $\vec{n}$ where $\vec{n}$ is the normal vector of the given plane, is -

(A) 3

(B) 5

(C) 15

(D) 25

5. Let $A = \left\{ \dfrac{x^2-p^2}{x^2-4x+p^2} \middle| x \in R,\ p \in R \right\}$ and

$B = \left\{ p \in I \middle| \dfrac{x^2-p^2}{x^2-4x+p^2} \in R \right\}$

If A is $(-\infty, \infty)$ then the value of $|B|$ is -

[Note : $|M|$ denotes the number of elements in set M]

(A) 2

(B) 3

(C) 4

(D) 5

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

6. Let $y = f(x)$ satisfies the differential equation $3x^3ydx - x^4dy = (y^2 + x^3y)dx$ where $x > 0$, $y > 0$ and $x \neq \frac{1}{2}$. If $y(1) = 1$ then identify which of the following statement(s) is(are) correct ?

(A) Area of the triangle formed by tangent and normal drawn to the curve at (1, 1) with the x-axis 1 sq. unit.

(B) If $f(x) = 1 + m(x - 1)$ has three distinct solutions then $m > 1$.

(C) If $f(x) = 1 + m(x - 1)$ has two distinct solutions then $m \in [0, 1)$.

(D) If $f(x) = 1 + m(x - 1)$ has two distinct solutions then $m \in (-\infty, 1)$.

7. Consider H : $\frac{x^2}{4} - \frac{y^2}{16} = 1$ and P : $y^2 = 4x$,

If M(x, y) lies inside the parabola from where two distinct tangents are drawn to the hyperbola H, then identify the correct statement(s) -

(A) If tangents touches two different branches of the hyperbola then area of the region in which M(x, y) lies is 2/3 sq. units.

(B) If tangents touches two different branches of the hyperbola then area of the region in which M(x, y) lies is 1/3 sq. units.

(C) If tangents touches two same branches of the hyperbola then area of the region in which M(x, y) $(0 \le x \le 2)$ lies is $\left(\frac{16\sqrt{2}-2}{3}\right)$ sq. units.

(D) If tangents touches two same branches of the hyperbola then area of the region in which M(x, y) $(0 \le x \le 2)$ lies is $\left(\frac{8\sqrt{2}-1}{3}\right)$ sq. units

8. If $f(x) = \max(x^2 - 4, 2 - |x|)$ and $g(x) = f(f(x))$ then identify which of the following statement(s) is(are) correct ?

(A) g(x) is non-derivable at 5 points.

(B) $\int_{-1}^{2} g(x)dx = \frac{1}{2}$

(C) The number of solution of the equation $g(x) = \frac{2015}{2014}$ is 8.

(D) g(x) is continuous for all $x \in R$.

9. If maximum length of the tangent drawn to the circle which touches the straight line $x + y = 4$ at (2,2) and also the positive co-ordinate axes, from the point $(x, \pm(4 - |x|))$ where $-6 \le x \le 4$ is $4\sqrt{a + b\sqrt{2}}$ then $(a + b)$ is

(Where $a, b \in Z$)

(A) 11

(B) 13

(C) 7

(D) 15

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.
Zero Marks : 0 In all other cases

---

1. If $\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} (\ell n|3\cos(x+\frac{\pi}{3}) + 5\cos x| + |3\sin^2 x - 6\sin x\cos x + \cos^2 x - 2|)dx = \pi\ell np + \sqrt{q}$

where p, q $\in$ Q then the value of $(p \cdot q)$ is equal to -

2. Let f : R$\longrightarrow$R be a function such that $f(x^5 + x - 2) = x^3 - 1\ \forall\ x \in R$ and g be the inverse function of f.

If $\ell = \frac{d}{dx}((x^3+1).g(f^3(g(f(g(x))))))\Big|_{x=7^{1/3}}$ and $m = \frac{d^2}{dx^2}\left(\frac{f(f(g(g(f(x)))))}{x^2}\right)\Big|_{x=-2}$ then the value of $\left(\ell(7^{-2/3}) + 8m\right)$ is -

3. If α is a non-real 7th root of 3 and $x = \alpha + \alpha^2$ then the value of $x^7 - 21[x^2(x+1) + x^2 + x + 1] + 15$ is -

4. Consider $f(x) = ||x-2| - |3x-1| + |2x+1||$ and

$g(x) = \left[|-x^2 + \frac{3}{2}x + 1|\right]$

[Note : [y] denotes greatest integer less than or equal to y.]

If m and n are the no. of points where $h(x) = f(x)\cdot g(x)$ is discontinuous and non-derivable respectively then $(m + n)$ is -

5. If $x + y + z + w = a^2 + b^2 - 4a + 4b$ where $a^2 + b^2 = 4$ and a, b $\in$ R then number of quadruplets (x, y, z, w) where x,y,z,w $\in$ N is -

6. Let the locus of the centre of the family of circles which intersects the parabola $x^2 - 2y + 3x = 4$ orthogonally at the point (1/2 , –9/8) be C and ℓ be the least distance of the curve C from $y = x - 1/x$ $(x < 0)$ then $(8\sqrt{5}\,\ell)$ is

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
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- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks : 0 If none of the options is chosen
Negative Marks : –1 In all other cases

1.

	Column-I		Column-II
(A)	If the point $(\alpha, \alpha^2 - 3)$, $\alpha \in R$ lies in acute angular region between the lines $7x - y - 15 = 0$ and $x - y + 3 = 0$ then largest integral value of α is	(P)	2
(B)	If A, B, C are three sets such that $n(A \cap B) = n(B \cap C) = n(C \cap A) = n(A \cap B \cap C) = 2$ then $n((A \times B) \cap (B \times C))$ is	(Q)	3
(C)	If the shortest distance between the lines formed by the intersection of the plane $2x - 3y + z = 1$ with the planes $x - y - z = 1$ and $x - y - z + 1 = 0$ is $2\sqrt{\frac{a}{b}}$ where a and b are prime numbers then $\lvert b - a \rvert$ is	(R)	4
(D)	Let 3 male and 4 female teachers be ready for a ramp walk in a line, if the number of ways in which even number of female teachers are there after every male teacher in the line, is 60 (K!) then K is	(S)	5
		(T)	6

2.

	Column-I	Column-II	
(A)	Let α and β be two positive integral roots of the equation $x^3 + (221 - 4m)x^2 + 220x + 4m = 0$ where $\alpha < \beta < 20$. Then $\left(\frac{\alpha + \beta + m}{19}\right)$ is less than or equal to -	(P)	0
(B)	If the set of values of p for which the equation $\sin^{-1}(2\|x\| - 1) = p(\|x\| - 1/2)$ has 6 distinct solutions is (a, b) then (a + b) is less than or equal to -	(Q)	2
(C)	Let $\bar{x}_\omega$ be the weighted mean of first n natural numbers when their weights are equal to their squares respectively then $\lim_{n\to\infty} \frac{\bar{x}_\omega}{n}$ is less than or equal to	(R)	3
(D)	If f is polynomial such that $f(f(f(x))) - 2f(f(x)) + 3x - 7 = 0\ \forall\ x \in R$ then $\frac{1}{2}\int_0^1 (f(x) - f^{-1}(x))dx$ is less than or equal to -	(S)	5
		(T)	8

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____	_____
Signature of the Candidate	Signature of the Invigilator

Space for Rough Work



English

(1001CJA101021230072)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 28)

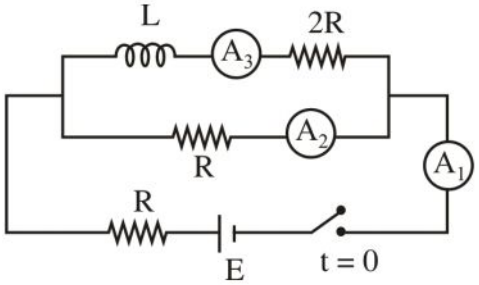
- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Figure shows a circuit consisting of an inductor and resistors connected with a cell. All ammeter are ideal. The switch was closed at t = 0. Choose the correct statement(s).



(A) Reading of A_1 is sum of reading of A_2 and A_3.

(B) Reading of A_2 will be $\frac{\varepsilon}{10R}\left(4+e^{-\frac{5Rt}{2L}}\right)$.

(C) Reading of A_3 will be $\frac{\varepsilon}{5R}\left(1-e^{-\frac{5Rt}{2L}}\right)$.

(D) In steady state reading of A_1 will be $\frac{\varepsilon}{2R}$.

2. The ground state and first excited state energies of hydrogen atom are -13.6 eV and -3.4 eV respectively. If potential energy in ground state is taken to be zero, then :-

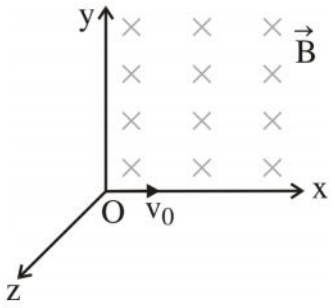
(A) Potential energy in the first excited state would be 20.4 eV.

(B) Total energy in the first excited state would be 23.8 eV.

(C) Kinetic energy in the first excited state would be 3.4 eV.

(D) Total energy in the ground state would be 13.6 eV.

3. A particle of mass m, charge q is projected from origin with velocity $v_0\hat{i}$ in a region where uniform magnetic field $\vec{B} = B_0(-\hat{k})$ exist. If it experience drag force equal to $\vec{F} = -K\vec{v}$, where K is a positive constant and $\vec{v}$ is velocity vector, then :-



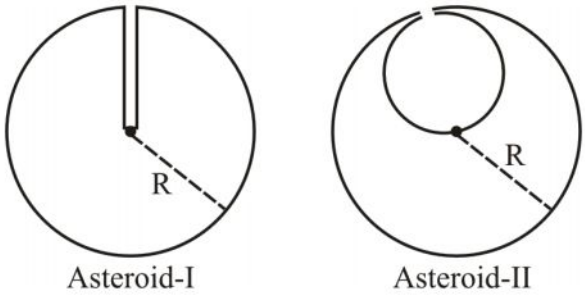
(A) Distance travelled by particle before its velocity is along the positive x-axis is $\frac{mv_0}{K}\left(1-e^{-\frac{2\pi K}{qB}}\right)$.

(B) Distance travelled by particle before its velocity is along the positive x-axis is $\frac{2mv_0}{K}\left(1-e^{-\frac{2\pi K}{qB}}\right)$.

(C) Speed of the particle decreases exponentially with time.

(D) Particle finally stops at origin.

4. Consider two solid spherical asteroid of uniform density of mass M and radius R. In one asteroid a tunnel of very small size of depth R is bored to the centre and in other asteroid a spherical cavity of radius $\frac{R}{2}$ is made as shown in the figure. Now, identical particles of mass m are dropped into cavities of both asteroids from top most point P. The force experienced by particles are F_I and F_{II} respectively in cavities of asteroid I and II, when they are x distance away from centre of asteroids. If the time taken by particles to reach the centre of asteroids is T_I and T_{II} respectively then



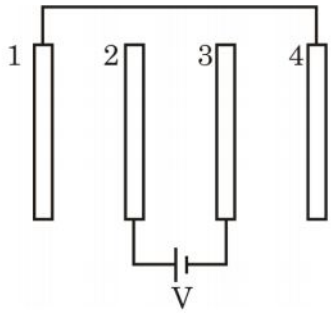
(A) The ratio of F_I/F_{II} is equal to $2x/R$.

(B) The ratio of F_I/F_{II} is equal to x/R.

(C) The ratio of T_I/T_{II} is equal to $\pi/4$.

(D) The ratio of T_I/T_{II} is equal to $\pi/2$.

5. Four identical conducting plates of area A each are connected as shown in circuit and the gap 'd' between adjacent plates is same. Initially charge on each plate is zero. Select the correct statements.



(A) Equivalent capacitance between the terminals of battery is $\left(\frac{3\epsilon_0 A}{2d}\right)$.

(B) Sum of charges on plate-1 and plate-4 is zero.

(C) The charge on right side of plate-2 is $\left(-\frac{\epsilon_0 AV}{d}\right)$.

(D) The charge on left side of plate-2 is zero.

6. Train is moving with constant speed along a circular track. If length of the train is one fourth of length of circular track then which of the following is/are correct options? (Assume that sound source is at engine).

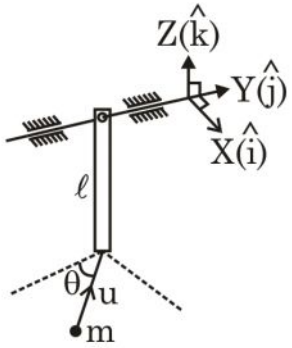
(A) Frequency observed by a passenger who is sitting in the middle of train (equidistance from front and rear end) will continuously increase.

(B) Frequency observed by a passenger who is sitting in the middle of train (equidistance from front and rear end) will remain constant but more than actual frequency.

(C) Frequency observed by passenger who is sitting in the middle of train (equidistance from front and rear end) will remains constant and equal to actual frequency.

(D) Wavelength observed by the person who is on the rear end of train is more than the actual wavelength of sound wave.

7. A particle of mass m moving in XY horizontal plane strikes the end of the vertical rod at an angle 37° with Y-axis. Rod has mass m and length ℓ and it is free to rotate about hinge along Y-axis. If particle strikes rod with speed u and sticks to it, find the **CORRECT** choice(s).



(A) Velocity of particle just after impact will be $-\frac{4u\hat{i}}{5}+\frac{9u\hat{j}}{5}$.

(B) Velocity of particle just after impact will be $-\frac{9u\hat{i}}{20}+\frac{4u\hat{j}}{10}$.

(C) Minimum speed so that rod completes full rotation about hinge is $\frac{10}{3}\sqrt{2g\ell}$.

(D) Angular momentum of system during collision is conserved about Y-axis.

SECTION-I (ii) : (Maximum Marks: 12)

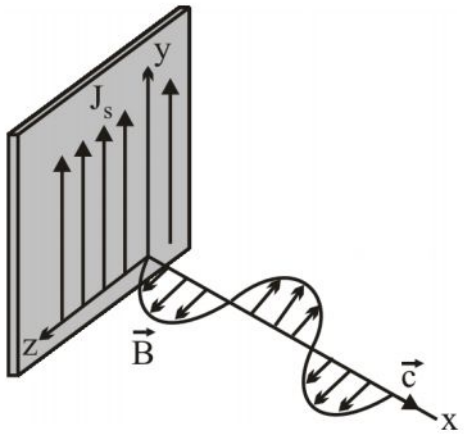
- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1	In all other cases.

Paragraph for Questions 8 and 9

A large, flat sheet carries a uniformly distributed electric current with current per unit width J. This current creates a magnetic field on both sides of the sheet, parallel to the sheet and perpendicular to the current, with magnitude $B = \frac{1}{2}\mu_0 J_s$. If the current is in the y direction and oscillates in time according to $J_{max}(\cos \omega t)\hat{j}$, the sheet radiates an electromagnetic wave. Figure shows such a wave emitted from one point on the sheet chosen to be the origin. Such electromagnetic waves are emitted from all points on the sheet. The magnetic field of the wave to the right of the sheet is described by the wave function

$$\vec{B} = \frac{1}{2}\mu_0 J_{max}[\cos(kx - \omega t)]\hat{k}$$



8. The electric field of wave to the right of the sheet is :-

(A) $\frac{1}{2}\mu_0 c J_{max}\cos(kx - \omega t)\hat{j}$

(B) $-\frac{1}{2}\mu_0 c J_{max}\cos(kx - \omega t)\hat{j}$

(C) $\frac{1}{2}\frac{\mu_0 J_{max}}{c}\cos(kx - \omega t)\hat{k}$

(D) $-\frac{1}{2}\frac{\mu_0 J_{max}}{c}\cos(kx - \omega t)\hat{k}$

9. If the intensity of radiation emitted by the sheet in each direction is 570 W/m^2, the maximum value of sinusoidal current density is :-

(A) 1.28 A/m

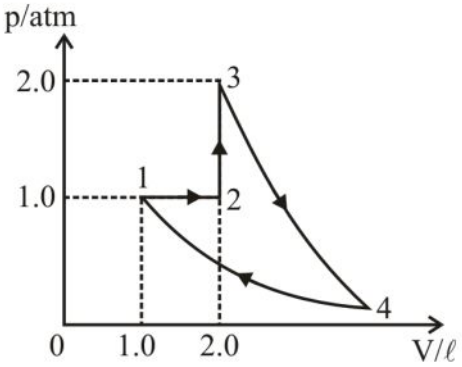
(B) 2.46 A/m

(C) 3.48 A/m

(D) 4.92 A/m

Paragraph for Questions 10 and 11

3.2g of oxygen starts from the initial state 1 (p_1 = 1.0 atm, V_1 = 1.0L), reaches state 2 (V_2 = 2.0L) through an isobaric process, and then passes through the isochoric process. The process reaches state 3 (p_3 = 2.0atm), and then reaches state 4 through an adiabatic process. The temperature of state 4 is exactly the same as the temperature of state 1. Finally, it returns to state 1 through an isothermal process, forming a cyclic process.



10. The efficiency of cyclic process is :-

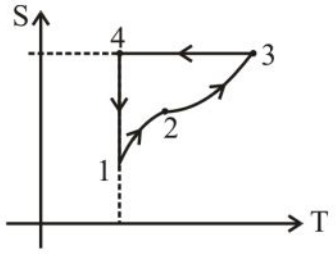
(A) $1-\frac{\ell n2}{2}$

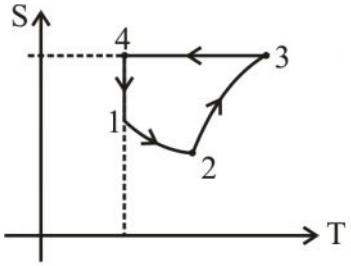
(B) $1-\frac{3\ell n2}{9}$

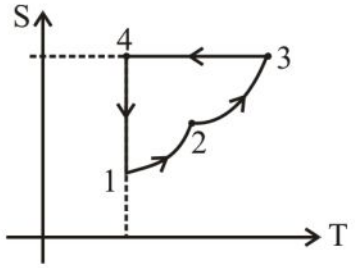
(C) $1-\frac{6\ell n2}{13}$

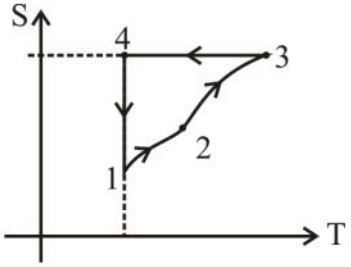
(D) $1-\frac{12\ell n2}{17}$

11. The variation of entropy with temperature is given by :-

(A)


(B)


(C)


(D)


SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.

If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

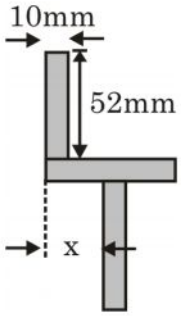
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

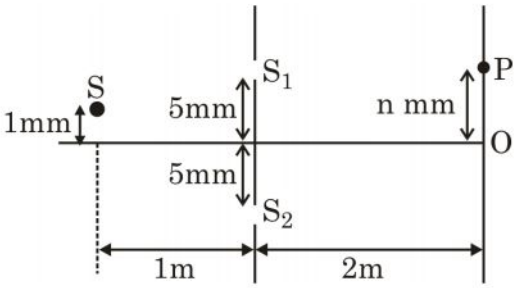
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1. Three identical uniform dominos are arranged as shown in the diagram. The possible values of x such that the dominoes are in stable equilibrium is $\alpha < x < \beta$. α, β and x are measured in mm. Then the value of $\alpha + \beta$ (in mm) is :-



2. A magnetic disc is moving on an air-cushioned table above a metal sheet. Due to the generated eddy currents there is a retarding force exerted on the disc, which is proportional to the speed of the disc. Moving above a sheet made of aluminum the disc stops after covering a distance of 30 cm, but when the sheet is made of copper the disc stops after covering only 20 cm. How much total distance (in cm) will the disc cover if first it travels above a piece of copper sheet of width 15 cm, and then continues its motion above an aluminum sheet? (The initial speeds of the disc are the same in all three cases.)

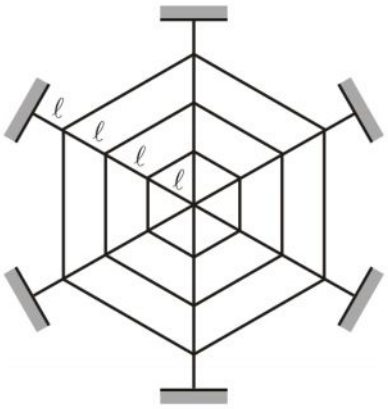
3. In the young"s double slit experiment a point source of $\lambda = 5000$ Å is placed slightly off the central axis as shown in the figure. If 70th order maximum is formed at P, the value of n is :-



4. A rocket is launched normal to the surface of the Earth, away from the Sun along the line joining the Sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2$ kms^{-1}. The minimum initial velocity (v_s) (in km/sec) required for the rocket to be able to leave the Sun-Earth system is closest to :- (Ignore the rotation and revolution of the Earth and the presence of any other planet)

5. Troposphere is the lowest layer of atmosphere, whose temperature decreases linearly with height (y) according to $T = T_0 - \alpha y$, where T_0 is temperature at Earth's surface and the constant α is called lapse rate of temperature. The average molar mass of normal atmospheric air is M_0 kg/mole and for the height of troposphere, the gravity field g is constant. The atmospheric pressure at ground is P_0 and gas constant is R. The atmospheric pressure as a function of height y will be $P(y) = P_0\left(1 - \frac{\alpha y}{T_0}\right)^{\frac{nM_0 g}{2\alpha R}}$, then value of n is :-

6. The web of negligible weight has the shape shown in the picture. The hexagons are regular and divide the radial fibers into equal parts of length ℓ. The ends of the web are fixed at the same height above the ground. At the beginning, the web is not sagged. When a spider of mass m moves to the center of the web, the web bends so that its center moves closer to the ground by a distance h. Determine the Young's modulus of elasticity Y of a spider fiber that has a diameter d. If the answer is $x \times 10^{17}$ Pa, fill the value of x. (Given : $\ell = 0.10$ m, m = 0.010 kg, h = 5.0 mm, d = 0.15 μm)



PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

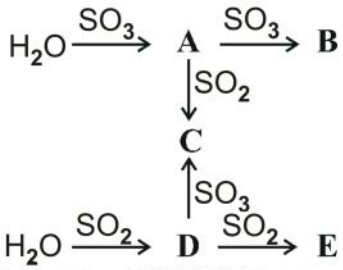
Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. The chain of reaction is completely hypothetical and based upon formula matching only.

Consider 1 molecule of all the compounds H_2O, SO_3, SO_2 , A, B, C, D, E in the given chain reaction



Select the **CORRECT** statement :

(A) When one oxygen atom is removed from formula of B then formula of C is formed

(B) **C** has S – S linkage in it's structure

(C) **E** has S – O – S linkage in it's structure.

(D) **C** is $H_2S_2O_6$ and E is $H_2S_2O_5$

2. Co-ordination entity — Wave length of light absorbed (nm)

$[CoCl(NH_3)_5]^{2+}$	P
$[Co(NH_3)_5(H_2O)]^{3+}$	Q
$[Co(NH_3)_6]^{3+}$	R
$[Co(CN)_6]^{3-}$	S
$[Cu(H_2O)_4]^{2+}$	T

Select the correct orders.

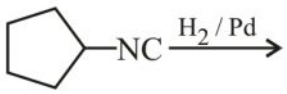
(A) P > R > Q

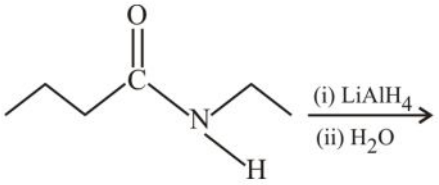
(B) T > P > Q > R > S

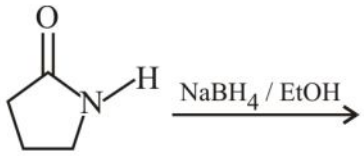
(C) P > Q > R

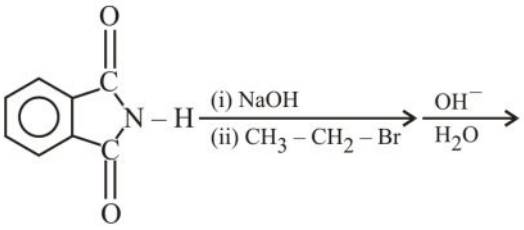
(D) T > Q > P > R > S

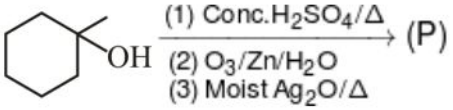
3. Which reactions/s can produce 2°-amines as major product?

(A) 

(B) 

(C) 

(D) 

4. 

Correct statement(s) for above reaction sequence :

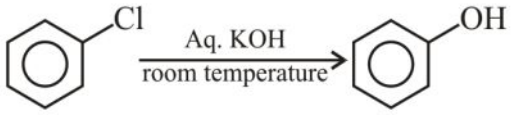
(A) Product (P) shows Tollen's test.

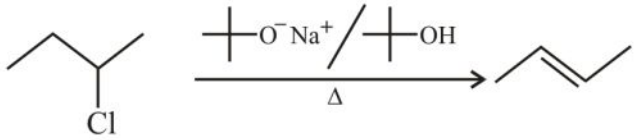
(B) Product (P) shows Iodoform test.

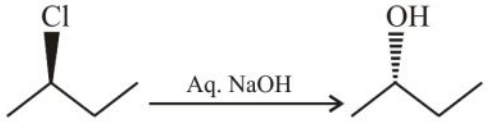
(C) Above reaction sequence involve elimination reaction.

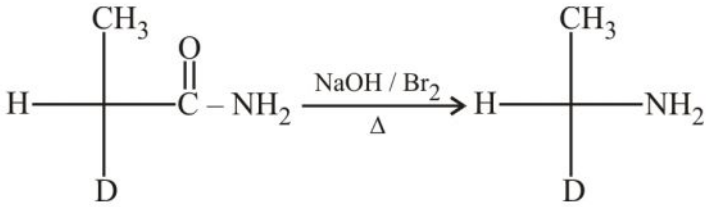
(D) Product (P) can show Lucas test

5. Which of the following reaction/s is/are correctly represented with major products?

(A) 

(B) 

(C) 

(D) 

6. $P_A = (235\ y - 125\ xy)$ mm of Hg. P_A is partial pressure of A, x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is the mole fraction of A in vapour phase, then :

(A) $P°_A = 110$

(B) $P°_B = 110$

(C) $P°_A = 235$

(D) $P°_B = 235$

7. The correct information(s) is/are :

(A) The minimum concentration of an electrolyte in millimoles per litre required to cause precipitation of a sol in two hours is called coagulation power

(B) Smaller the coagulation value, higher will be the coagulation power of that ion.

(C) Lyophilic sols may be coagulated by adding an electrolyte or a suitable solvent.

(D) Lyophobic colloids used to protect lyophilic colloids are called protective colloids.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3 If ONLY the correct option is chosen.
Zero Marks	:	0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1 In all other cases.

Paragraph for Questions 8 and 9

The electrochemical cell shown below is a concentration cell.

$M \mid M^{2+}$ (saturated solution of a sparingly soluble salt, MX_2) $\mid\mid M^{2+}$ (0.001 mol dm^{-3}) $\mid M$

The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059V.

8. The value of ΔG (kJ mol^{-1}) for the given cell $M(s) + M^{2+}(aq) \longrightarrow M(s) + M^{2+}(aq)$ is (take F = 96500 C mol^{-1})

(A) –5.7

(B) 5.7

(C) 11.4

(D) –11.4

9. The solubility product (K_{sp} ; mol^3 dm^{-9}) of MX_2 at 298 K based on the information available for the given concentration cell is $x \times 10^{-15}$ (take $2.303 \times R \times 298 | F = 0.059$ V). The value of x is ?

(A) 1×10^{-15}

(B) 4×10^{-15}

(C) 1×10^{-12}

(D) 4×10^{-12}

Paragraph for Questions 10 and 11

If values of quantum numbers can be considered as follows.

$n = 1, 2, 3 \ldots\ldots \infty$

$\ell = 0$ to $(n + 1)$

$m = -\ell$ to $+\ell$ (integral value including zero)

$s = -\frac{1}{4}, -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{4}$

10. According to above consideration, the maximum number of e^- associated to 2nd shell is :

(A) 64

(B) 32

(C) 16

(D) 8

11. According to above consideration, the maximum number of e^- associated to a p-orbital is :

(A) 2

(B) 4

(C) 6

(D) 12

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.

If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

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1. How many of the following cations gives ppt with H_2S in acidic solution :-
Zn^{2+} , Cu^{2+} , Ni^{2+} , Sr^{2+} , Al^{3+} , Bi^{3+} , Sn^{2+} , Pb^{2+}

2. Among the following statements

(i) Ag_2SO_3 is insoluble in dilute HNO_3.

(ii) Nitrogen dioxide (NO_2) is absorbed by ferrous sulphate solution forming brown solution.

(iii) Heavy metal chlorides like AgCl, $HgCl_2$. etc. do not respond to chromyl chloride test.

(iv) The group reagent for III group basic radicals is NaOH.

(v) Green colour of alkaline tetrahydroxidochromate (III) becomes yellow on adding H_2O_2.

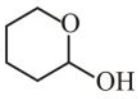
(vi) Titan yellow solution gives blue colouration/ppt. with a acidic solution containing Mg^{2+} ions

(vii) Solution of nitrite is decomposed by sulphamic acid.

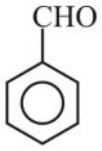
(viii) Fe^{2+} ions give brown colour precipitate with $[Fe(CN)_6]^{3-}$ ions solution.

(ix) Green precipitate of $Cr(OH)_3$ is soluble in excess Na_2O_2.

If "**a**" is total number of correct while "**b**" is total number of incorrect statements find $\frac{a^2 + b^2}{3}$.

3. (i)  (ii) $CH_3-\overset{\overset{O}{\|}}{C}-H$ (iii) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

(iv) $CH_3-\overset{\overset{O}{\|}}{C}-\underset{\underset{OH}{|}}{CH}-CH_3$ (v) $CH_3-C\equiv C-CH_3$

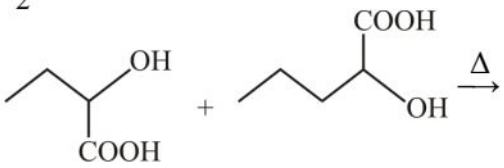
(vi) $H-\overset{\overset{O}{\|}}{C}-OH$ (vii)  (viii) CH_3-CH_2-OH (ix) $CH_3-\overset{\overset{O}{\|}}{C}-H$

Number of compounds that gives Tollen's test = x.

Number of compounds that gives Fehling's test = y.

Then x + y = _____

4. 'X' is the number of possible products (including stereoisomers) in the following reaction, then value of $\frac{X}{2}$ is



5. Fixed amount of an ideal gas contained in a sealed regid vessel (V = 24.6 litre) at 1. 0 bar is heated reversibly from 27ºC to 127º C. Determine change in Gibb's energy (in J), if entropy of gas $S = 10 + 10^{-2}\,T$ (J/K)

6. The solubility of a salt (AB) of weak acid at pH = 3 is $\sqrt{Y} \times 10^{-3}$ mol L^{-1}. The value of Y is___.
(Given that the value of solubility product of AB (K_{sp}) = 2×10^{-10} and the value of ionization constant of HB(K_a) = 1×10^{-8})

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. Let A, B, C be 3 sets of complex numbers as defined below :

$A = \{z : |z + 1| \le 2 + Re(z)\}$, $B = \{z : |z - 1| \ge 1\}$ and $C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$

Then which of the following are correct :

(A) Area of the region bounded by $A \cap B \cap C$ is $\sqrt{3}$

(B) Area of the region bounded by $A \cap B \cap C$ is $2\sqrt{3}$

(C) The real part of the complex number in region $A \cap B \cap C$ having maximum amplitude and modulus is $-\frac{3}{2}$

(D) The number of point having integral coordinates in region $A \cap B \cap C$ is 7

2. If A is the area enclosed between the curves $y = \ell n(x + e)$, $x = \ell n\left(\frac{1}{y}\right)$ and x-axis then :

(A) A is prime number

(B) A is composite number

(C) A is irrational number

(D) A^2 is even integer

3. The solution of the differential equation $3\frac{dx}{dy} = \frac{x}{x^3 - y}$ is $x^{\ell} = mx^n y + c$, then which of the following options are correct :

(A) $\ell + m + n = 11$

(B) $\ell + n = 9$

(C) $\ell + 2m = 10$

(D) $m + n = 4$

4. The value of $\sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$ is less than :

(A) $\frac{11}{7}$

(B) $\frac{11}{8}$

(C) $\frac{11}{9}$

(D) $\frac{11}{10}$

5. Tangent drawn at point P(2, –5) of a parabola intersects its tangent at vertex at (1, –3). If (8, 3) is a point on SP, where S is focus of the parabola then

(A) focus of the parabola is (5, –1)

(B) Directrix of the parabola is $x - y - 2 = 0$

(C) Slope of normal to parabola at other end of focal chord drawn from point P is –2

(D) Circle drawn on PS as diameter touches the straight line $x - y - 4 = 0$

6. Consider the equation $(x^2 - 4x + 6)^2 - (k + 2)(x^2 - 4x + 6) + k + 5 = 0$ ($k \in R$). Which of the following are correct :

(A) there is no integral value of k for which given equation has exactly 4 real solutions.

(B) there is only one integral value of k for which given equation has exactly 3 real solutions.

(C) there is only one integral value of k for which given equation has exactly one real solution.

(D) there are only 3 positive integral values of k for which given equation has no real solution.

7. Consider the limit $\lim_{x \to 0} \frac{a\tan x + b\tan^{-1}x + c\sin^{-1}x}{x^5} = 3$, where $a, b, c \in R$, then correct option/s is/are

(A) $a = 30$

(B) $b = 10$

(C) $c = -40$

(D) $b = -5$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1	In all other cases.

Paragraph for Questions 8 and 9

Let $f(x) = \begin{cases} x^2 - 3x + 2 & 0 \leqslant x \leqslant 3 \\ |x - 5| & 3 < x \leqslant 8 \\ \log_2(12 - x) & 8 < x \leqslant 11 \end{cases}$

8. If $f(x) = k$ has 5 distinct solutions then sum of integral values of k is

(A) 1

(B) 3

(C) 5

(D) 7

9. Let ℓ = maximum number of distinct roots of $|f(x)| = k$ and $m = \lim\limits_{x \to 3} \left[f(3)^{f(x)} \right]$, then $\ell + m$ is (Where [.] denotes greatest integer function)

(A) 11

(B) 9

(C) 7

(D) 5

Paragraph for Questions 10 and 11

Let $\frac{d}{dx}(f(x)) = \frac{e^{\sin x}}{x}$, $x > 0$, then answer the following :

10. If $\int_1^4 \frac{2e^{\sin x^2}}{x}\,dx = f(k) - f(1)$ and $xf''(x) + f'(x) = g(x)$, then $k + \left[g\left(\frac{\pi}{2}\right)\right]$ is

(A) 9

(B) 16

(C) 27

(D) 36

11. Let $I = \int_0^{\pi/2} x\cos x\, f'(x)\,dx = e^{\lambda} - 1$ and $xf'(x) = T(x)$, then $[\lambda] + T\left(\frac{\pi}{2}\right)$ is equal to : (Where [.] represents G.I.F.)

(A) $1 + e$

(B) 3

(C) 2

(D) 4

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.

If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. If k is the number of four digit natural numbers, such that repetition of digits is not allowed and absolute difference between its first and last digit is 2 then $\frac{k}{480}$ is :

2. If a is the minimum value of x_1 and b is the maximum value of x_2 such that $\int_{x_1}^{x_2} (x^4 - 2x^2)\,dx$ is minimum then the area of region enclosed between the curves $y = x^4 - 2x^2$, the x-axis and the ordinates $x = a$, $x = b$ is :

3. If a variable circle touches the straight line $y = x$ and cuts the chord of length $2\sqrt{2}$ from $x + y = 0$ then the length of latus rectum of the locus of the centers of the circles is :

4. If the value of $\lim\limits_{n\to\infty} n\left(\frac{1}{3n^2 + 8n + 4} + \frac{1}{3n^2 + 16n + 16} + \frac{1}{3n^2 + 24n + 36} + \ldots\ldots\right)$ is $\frac{1}{4}\ell n\left(\frac{n}{k}\right)$ where $\gcd(n, k) = 1$ then $\frac{n}{k}$ is :

5. For two data sets each of the size 5, variance are given to be 4 and 5 respectively and mean are 2, 4 respectively, the variance of combined data is equal to

6. Let A is a 2×2 matrix such that $AA^T = I$, where I is identity matrix then the maximum value of the sum of the absolute values of entries of matrix A is K then find [K] where [.] represents greatest integer function :

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	C	C	C	B	B	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A,B,C	A,B	A,D	A,D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	16	2	9	8	6	320
SECTION-IV	Q.	1	2				
	A.	A->Q,B->PR,C->S,D->T	A->Q,B->S,C->R,D->P				

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	D	D	D	A	A	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A	A,B,C,D	A,C,D	B,D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	5	1	9	5	2
SECTION-IV	Q.	1	2				
	A.	A->PQ,B->PST,C->PR,D->P	A->P,B->PQS,C->PR,D->QST				

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	B	D	A	D	A	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A,B,D	A,C	A,D	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	140	255	6	5	1365	33
SECTION-IV	Q.	1	2				
	A.	A->P,B->R,C->T,D->R	A->ST,B->ST,C->QRST,D->PQRST				

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

The decay law will remain same even in the train. The velocities of the α-particle and the recoiling nucleus will be same on the ground and in the train with respect to train.

2. Ans (C)

Centre point is the unstable equilibrium position where potential energy is maximum.

3. Ans (C)

Gravitational field intensity due to ring,

$$I_g = \frac{\sqrt{3}GMa}{(a^2+3a^2)^{3/2}} = \frac{\sqrt{3}GM}{8a^2}$$

$$|F_{sphere}| = \frac{\sqrt{3}GMm}{8a^2}$$

$$|F_{ring}| = \frac{\sqrt{3}GMm}{8a^2}$$

$$a_{ring} = \frac{\sqrt{3}GM}{8a^2}$$

4. Ans (B)

$mg = kx_{eq} \Rightarrow 4 \times 80 \times 10 = k \cdot \left(\frac{1.6}{100}\right)$

$k = 2 \times 10^5$ N/m

$T = 2\pi\sqrt{\frac{m}{k}} = \frac{\Delta x}{v}$

$2\pi\sqrt{\frac{320}{2 \times 10^5}} = \frac{8}{v} = 2\pi\sqrt{16 \times 10^{-4}}$

$\frac{8}{v} = 2\pi \times 4 \times 10^{-2}$

$v = \frac{8}{8\pi \times 10^{-2}} = \frac{\pi}{\pi^2} \times 100 = 10\pi$

5. Ans (B)

Case-I :

$f_A = f_C - 8 \quad f_B = f_A (1 + 0.06)$

$f_B = f_C + 7$

$f_B - f_A = 15$

$f_A (0.06) = 15$

$f_A = 250$ Hz

$f_B = 265$ Hz

$f_C = 258$ Hz

Case-II :

$f_A = f_C + 8$ (Not possible)

$f_B = f_C + 7$

Case-III

$f_A = f_C - 8$

$f_B = f_C - 7$

$f_B - f_A = 1$

$f_A (1 + 0.06) - f_A = 1$

$f_A \times 0.06 = 1$

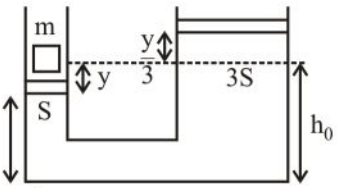
$f_A = 16.67$ (Not in audible range)

Hence only case-I is possible.

PART-1 : PHYSICS

SECTION-I (ii)

6. Ans (A,B,C)



Assuming that the wires have a resistance per unit length α.

$V_0 = I_0 (2\alpha h_0 + R_A)$

$P_0 + \frac{mg}{S} + \rho g (h_0 - y) = P_0 + \rho g \left(h_0 + \frac{y}{3}\right)$

$\frac{mg}{S} = \rho g \left(\frac{y}{3} + y\right) = \rho g \cdot \frac{4y}{3} = \frac{4}{3}\rho g y$

$y = \frac{3}{4}\frac{m}{\rho S}$

$V_0 = I_1 \left(R_A + \alpha (h_0 - y) + \alpha \left(h_0 + \frac{y}{3}\right)\right)$

$= I_1 \left(R_A + 2\alpha h_0 - \frac{2\alpha y}{3}\right)$

$= I_1 \left(R_A + 2\alpha h_0 - \frac{2\alpha}{3} \cdot \frac{3}{4}\frac{m}{\rho S}\right)$

$V_0 = I_1 \left(R_A + 2\alpha h_0 - \frac{\alpha m}{2\rho S}\right)$ .... (i)

$V_0 = I_2 \left(R_A + 2\alpha h_0 + \frac{\alpha m}{2\rho S}\right)$ ... (ii)

$2.31 \left(R_A + 2\alpha h_0 - \frac{\alpha m}{2\rho S}\right) = 2.1 \left(R_A + 2\alpha h_0 + \frac{\alpha m}{2\rho S}\right)$

$0.21 (R_A + 2\alpha h_0) = (4.41) \frac{\alpha m}{2\rho S}$

$R_A + 2\alpha h_0 = 21 \cdot \frac{\alpha m}{2\rho S}$ ... (iii)

$V_0 = I_2 \left(\frac{21\alpha m}{2\rho S} + \frac{\alpha m}{2\rho S}\right) = I_2 \cdot \frac{22\alpha m}{2\rho S} = \frac{11\alpha m}{\rho S} \cdot I_2$

$46.2 = \frac{11\alpha \times 1}{10^3 \times 100 \times 10^{-4}} \times 2.1$

$\alpha = 20$

$$R_A + 2 \times 20 \times \frac{1}{2} = \frac{21 \times 20 \times 1}{2 \times 10^3 \times 100 \times 10^{-4}}$$

$$R_A + 20 = 21$$

$$R_A = 1\Omega$$

when no load is present,

$$V = I_0 (2\alpha h_0 + R_A)$$

$$46.2 = I_0 \left(2 \times 20 \times \frac{1}{2} + 1\right)$$

$$I_0 = 2.2A$$

For maximum current, $Sh_0 + 3Sh_0 = 3Sh_1$

$$h_1 = \frac{4}{3}h_0$$

$$46.2 = I_{max} \left(R_A + \frac{4}{3}\alpha h_0\right)$$

$$= I_{max} \left(1 + \frac{4}{3} \times 20 \times \frac{1}{2}\right) = I_{max} \left(1 + \frac{4}{3} \times 10\right)$$

$$I_{max} = \frac{46.2}{14.33} = 3.22A$$

For minimum current, $Sh_0 + 3Sh_0 = Sh_2$

$$h_2 = 4h_0$$

$$I_{min} = \frac{46.2}{R_A + \alpha(4h_0)}$$

$$= \frac{46.2}{1 + 20 \times 4 \times \frac{1}{2}} = \frac{46.2}{41} = 1.126A$$

7. Ans (A,B)

Since the angles in a triangle formed by tigers remain constant, at any moment in time the triangle formed by tigers is similar to the original one. It follows from this that all tigers will meet at the same time.

Let's find the rate of change of distance AB. Since the speed modules of the tigers and the angles between them remain constant, the required speed is equal v.

The time before the first and second tigers meet is $t = L/v$.

Let us express the rates of change of distances BC and AC:

$$v_{BC} = v_2 + \frac{v_3}{\sqrt{2}},$$
$$v_{AC} = v_3 + \frac{v_1}{\sqrt{2}}$$

Since the angles in triangle ABC remain constant, the relationships between its sides also remain constant. This means that the rates of change of the sides of the triangle are proportional to them

$$\frac{v_{AB}}{AB} = \frac{v_{BC}}{BC} = \frac{v_{AC}}{AC}.$$

$$\frac{v}{L} = \frac{v_3 + \frac{v}{\sqrt{2}}}{L\sqrt{2}} \Rightarrow v_3 = \frac{v}{\sqrt{2}}.$$

$$\frac{v}{L} = \frac{v_2 + \frac{v}{2}}{L} \Rightarrow v_2 = \frac{v}{2}.$$

Since the velocity modulus of the second tiger is always half the velocity modulus of the first tiger, and the directions of velocities at any moment are mutually perpendicular, the trajectories of their movement will be similar to each other with a similarity coefficient of 2 : 1. It follows that the displacement vectors will also be mutually perpendicular and differ in magnitude by a factor of two.

This can also be proven strictly mathematically. Let's consider a small moment of time dt. Let the angle between the speed of the first tiger and the x-axis be equal to a, then the angle between the speed of the second tiger and the x-axis will be equal to a + 90°. Let us write down the changes in the coordinates of tigers in a short time:

$dx_1 = v\cos\alpha \cdot dt$,

$dy_1 = v\sin\alpha \cdot dt$,

$dx_2 = \frac{v}{2}\cos(\alpha + 90°) \cdot dt = -\frac{v}{2}\sin\alpha \cdot dt$,

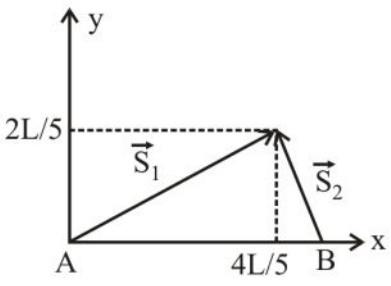
$dy_2 = \frac{v}{2}\sin(\alpha + 90°) = \frac{v}{2}\cos\alpha \cdot dt$.

From the resulting equations it is clear that for any moment of time $dx_1 = 2dy_2$, $dy_1 = -2dx_2$, this means that the vectors of total displacements also differ by a factor of two in absolute value and are mutually perpendicular

Since the tigers met: $\vec{S}_1 = \overrightarrow{AB} + \vec{S}_2$.

Then from the right triangle we find

$(x_B, y_B) = \left(\frac{4L}{5}, \frac{2L}{5}\right)$.



8. Ans (A,D)

According to Fermat's principle, all rays of light incident as parallel light and converging at point F should have equal optical path lengths, that is, the optical path lengths of the edgemost light BF and any light NMF should be equal [M is any point on the convex surface, its coordinates is M(x,y)], from this, the equation of the convex surface can be determined, and other problems can also be easily solved.

Taking the Oxy coordinate as shown in Figure 1, based on the isopticality of light BF and NMF, we get

$L = nx + \sqrt{(f-x)^2 + y^2} = \sqrt{f^2 + R^2}$

Squaring the above formula and sorting it out a little, we can get the equation that the coordinates (x, y) of any point M (x, y) on the convex surface of the lens should satisfy

$$(n^2-1)\left(x - \frac{n\sqrt{f^2+R^2} - f}{n^2-1}\right)^2 - y^2 = \frac{\left(nf - \sqrt{f^2+R^2}\right)^2}{n^2-1}$$

$$x_0 = \frac{n\sqrt{f^2+R^2} - f}{n^2-1}$$

$$a = \frac{nf - \sqrt{f^2+R^2}}{\sqrt{n^2-1}}$$

$$(n^2-1)(x-x_0)^2 - y^2 = a^2$$

This is the equation of the hyperbola. Due to rotational symmetry, the convex surface of the lens should be a rotational hyperboloid.

The position X_A of the lens vertex A should satisfy

$$(n^2-1)(x_A - x_0)^2 = a^2$$

$$x_A = x_0 + \frac{a}{\sqrt{n^2-1}} = \frac{\sqrt{f^2+R^2} + f}{n+1}$$

Since point F is outside the lens, that is $x_A \le f$

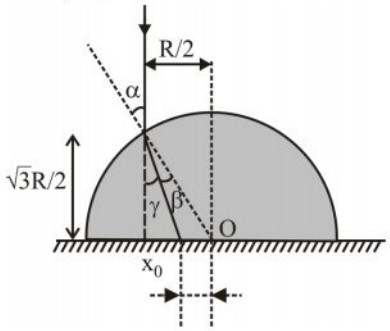
This is a restriction on R, that is $\frac{\sqrt{f^2+R^2} + f}{n+1} \leqslant f$

$R \leqslant \sqrt{n^2-1}\, f$

In the extreme case, that is, when

$R = \sqrt{n^2-1}\, f$

9. Ans (A,D)



(a) Let's find the point on the table where the beam hits at the very first moment t = 0. Consider Figure, it is easy to find that the angle of incidence of the beam on the hemisphere is $\alpha = 30°$, i.e. $\sin\alpha = \frac{1}{2}$.

According to Snell's law of refraction

$$\sin\beta = \frac{\sin\alpha}{n} = \frac{1}{2n} \Rightarrow \tan\beta = \frac{1}{\sqrt{4n^2-1}}$$

From the triangle with angle γ in Figure it is easy to find its horizontal leg $\frac{\sqrt{3}R}{2}\tan\gamma$, which means the point of incidence of the beam on the table: it is located at a distance x_0 from the center of hemisphere O:

$$x_0 = \frac{R}{2} - \frac{\sqrt{3}}{2}R\tan\gamma$$

$$\tan\gamma = \tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta} = \frac{\sqrt{4n^2-1}-\sqrt{3}}{\sqrt{12n^2-3}+1}$$

$$x_0 = \frac{R}{2}\left(1 - \frac{\sqrt{12n^2-3}-3}{\sqrt{12n^2-3}+1}\right) = \frac{2R}{\sqrt{12n^2-3}+1}$$

(b) At time t, the base in contact with the table is a circle with radius r. Within a time interval dt, the amount of heat absorbed is

$$dQ = K\pi r^2 dt$$

for some constant K. The volume of the ice melt is

$$dQ = Ldm = L\rho\pi r^2 dz$$

Where L and ρ are the latent heat and density of the ice respectively. Therefore we have

$$K\pi r^2 dt = -L\rho\pi r^2 dz$$

$$\Rightarrow \frac{dz}{dt} = -\frac{K}{L\rho} \text{ is a constant}$$

i.e. the melting uniformly reduces the thickness of the ice, we have

$$z(t) = R\left(1 - \frac{t}{T_0}\right)$$

(c) At the end, when the base of the ice has radius $r \leqslant \frac{R}{2}$, the beam will obviously hit the table vertically (i.e. $x(t) = \frac{R}{2}$). It corresponds to the case when the thickness of the ice reduces to z(t)

$$= \left(1 - \frac{\sqrt{3}}{2}\right)R.$$ It will happen at the time $t = \frac{\sqrt{3}}{2}T_0$. Therefore, for $t > \frac{\sqrt{3}}{2}T_0$,

we have $x(t) = \frac{R}{2}$

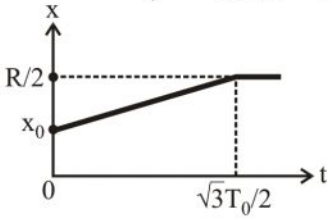
To get x(t) for $t < \frac{\sqrt{3}}{2}T_0$, we can consider the similar triangles.

$$\frac{\frac{R}{2} - x_0}{\frac{R}{2} - x(t)} = \frac{\frac{\sqrt{3}}{2}R}{R\left(\frac{\sqrt{3}}{2} - \frac{t}{T_0}\right)}$$

$$\Rightarrow \left(\frac{R}{2} - x_0\right)\left(\frac{\sqrt{3}}{2} - \frac{t}{T_0}\right) = \frac{\sqrt{3}}{2}\left(\frac{R}{2} - x\right)$$

$$\Rightarrow x(t) = \frac{R}{2} - \left(\frac{R}{2} - x_0\right)\left(1 - \frac{2}{\sqrt{3}}\frac{t}{T_0}\right)$$

$$= \frac{R}{2} - \frac{R}{2}\left(1 - \frac{4}{\sqrt{12n^2-3}\,\backslash;+1}\right)\left(1 - \frac{2}{\sqrt{3}}\frac{t}{T_0}\right)$$

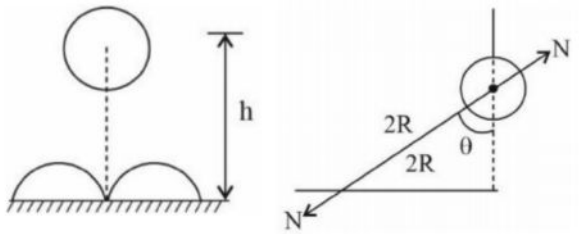


PART-1 : PHYSICS

SECTION-II

3. Ans (9)

$$\sin\theta = \frac{1}{\sqrt{3}}$$



$$\therefore \cos\theta = \sqrt{\frac{2}{3}} \therefore \int 3N\cos\theta dt = mv_0$$

$$\int N\sin\theta dt = \frac{m}{2}v$$

$$\therefore 3\cot\theta = \frac{2v_0}{v}$$

$$\therefore v = \frac{2}{3\sqrt{2}}v_0 = \frac{\sqrt{2}}{3}v_0 \quad \therefore p = \frac{\frac{1}{2}mv_0^2}{\frac{1}{2}\frac{m}{2}\cdot\frac{2v_0^2}{9}} = 9$$

4. Ans (8)

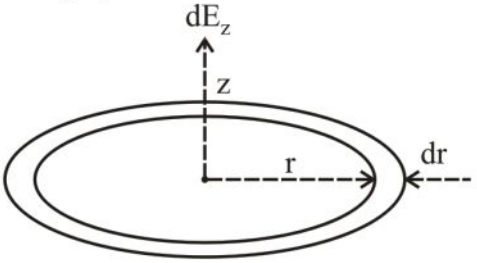
Height of water column $= \frac{20}{5} = 4\text{cm}$

Mass of water = 20 gm

By work energy theorem

$$N^{ext} = \Delta U^{Piston} + \Delta U^{Water} + KE^{Water} = 8\times10^{-2}\text{ J}$$

5. Ans (6)



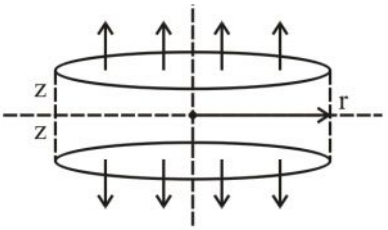
for $z << r$

$dq = \sigma(2\pi r)\,dr$

$$E_z = k\sigma(2\pi)z\int_{r_1}^{r_2}\frac{dr}{r^2}$$

$$= k\sigma(2\pi)z\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

In order to calculate radial component of $\vec{E}$ in the plane of ring for $r < R_1$



$$E_z(2\pi r^2) + E_r(2\pi(2z)r) = 0$$

$$E_z(2\pi r^2) = -E_r(4\pi rz)$$

$$E_r = -\frac{E_z r}{2z}$$

$$= -\frac{r}{2z}\cdot k\sigma(2\pi)z\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$$E_r = -\frac{\sigma}{4\epsilon_0}\cdot r\left(\frac{r_2-r_1}{r_1r_2}\right)$$

at r = R,

$$\frac{kQ}{R^2} = \frac{\sigma R}{4\epsilon_0}\cdot R\left(\frac{r_2-r_1}{r_1r_2}\right)$$

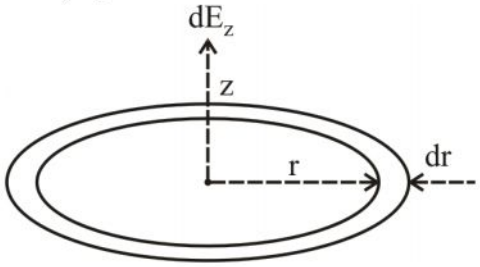
$$\frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R^2} = \frac{\sigma R^2}{4\epsilon_0}\cdot\left(\frac{r_2-r_1}{r_1r_2}\right)$$

$$\sigma = \frac{Q}{\pi R^3}\left(\frac{r_1r_2}{r_2-r_1}\right)$$

for R = 1cm, $r_1 = 2m_1$, $r_2 = 3m$, $Q = \pi\times10^{-6}$ C

$\sigma = 6\text{C/m}^2$

5. **Ans (6)**



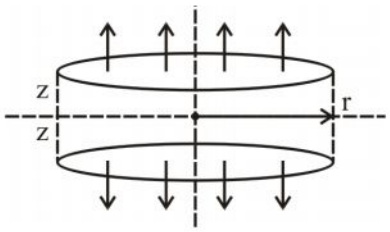
for $z << r$

$dq = \sigma(2\pi r)\, dr$

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$$E_r = -\frac{E_z r}{2z}$$

$$= -\frac{r}{2z}\cdot k\sigma(2\pi)z\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$$E_r = -\frac{\sigma}{4\epsilon_0}\cdot r\left(\frac{r_2-r_1}{r_1r_2}\right)$$

at $r = R$,

$$\frac{kQ}{R^2} = \frac{\sigma R}{4\epsilon_0}\cdot R\left(\frac{r_2-r_1}{r_1r_2}\right)$$

$$\frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R^2} = \frac{\sigma R^2}{4\epsilon_0}\cdot\left(\frac{r_2-r_1}{r_1r_2}\right)$$

$$\sigma = \frac{Q}{\pi R^3}\left(\frac{r_1r_2}{r_2-r_1}\right)$$

for $R = 1cm$, $r_1 = 2m_1$, $r_2 = 3m$, $Q = \pi \times 10^{-6}\,C$

$\sigma = 6C/m^2$

6. **Ans (320)**

$$\frac{mgR - \frac{mgR^2}{R+h} - mgh}{mgR - \frac{mgR^2}{R+h}} = \frac{5}{100}$$

$$\frac{Rh - Rh - h^2}{Rh} = \frac{5}{100}$$

$$\frac{h}{R} = \frac{5}{100}$$

$$k = \frac{5}{100}\times 6400 = 320km$$

PART-1 : PHYSICS

SECTION-IV

1. **Ans (A->Q,B->PR,C->S,D->T)**

Suppose the plank is displaced by x and at that instant its speed is v then speed of COM of cylinder v/2.The total energy is

$$E_T = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 + \frac{1}{2}\cdot\frac{2mR^2}{2}\left(\frac{v/2}{R}\right)^2$$

$$= \left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right)mv^2 + \frac{1}{2}kx^2$$

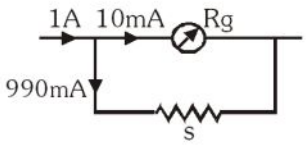
$$= \frac{4+2+1}{8}mv^2 + \frac{1}{2}kx^2 \text{ or } E_T = \frac{7}{8}mv^2 + \frac{1}{2}kx^2$$

Also, $E_T = \frac{1}{2}kx_0^2$

$\frac{dE_T}{dt} = 0$, after calculating we get $a = -\frac{4k}{7m}x$

Comparing with $a = -\omega^2 x$, we get $\omega = \sqrt{\frac{4k}{7m}}$

2. Ans (A->Q,B->S,C->R,D->P)



For (A)

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k} = \frac{1}{x^2}\hat{i} + \frac{1}{y^2}\hat{j} + \frac{1}{z^2}\hat{k}$$

$$\Rightarrow |\vec{E}| = |\hat{i} + \hat{j} + \hat{k}| = \sqrt{3}$$

For (B) $I = \frac{dQ}{dt} = 4 - 4t$,current will be zero

at t=1s

Total heat produced

$$\int_0^1 I^2 R dt = \int_0^1 (4-4t)^2 (6)\, dt = (16 \times 6)\left[\frac{(1-t)^2}{-3}\right]_0^1 = 32$$

For(C) (10 mA) (Rg) = (990 mA) (s)

$$\Rightarrow s = \frac{0.99}{99} = 0.01\Omega$$

For(D) $U = -\vec{p}.\vec{E}$ Here $\vec{p} = \pm q d\hat{i}$

$$= \pm (10 \times 10^{-3})(8 \times 10^{-2}) = \pm (8 \times 10^{-4})$$

$$\Rightarrow U = \pm(8 \times 10^{-4})(500) = \pm 0.4$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (D)

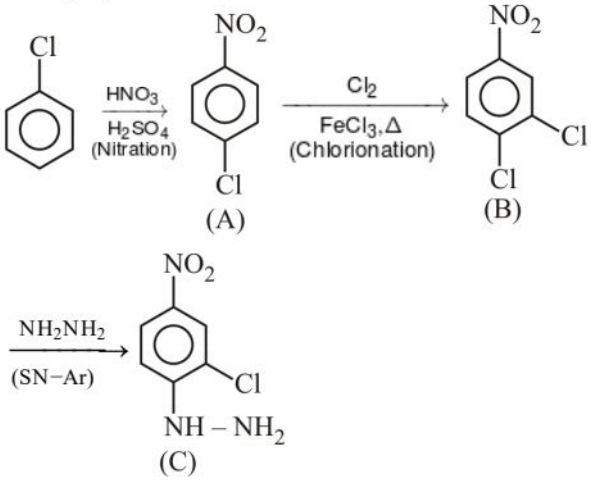
$$\frac{r_A}{r_B} = \frac{2}{1} = \sqrt{\frac{M_B}{M_A}}$$

$$\therefore \frac{M_B}{M_A} = \frac{4}{1}$$

$$\frac{(U_{rms})_A}{(U_{rms})_B} = \sqrt{\frac{T_A}{M_A} \times \frac{M_B}{T_B}} = \sqrt{\frac{2}{1} \times \frac{4}{1}}$$

$$= \frac{2\sqrt{2}}{1}$$

5. Ans (A)



PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (A,C,D)

Gibb's free energy of formation of all the elements in their natural standard state is arbitrarily assumed to be zero.

PART-2 : CHEMISTRY

SECTION-II

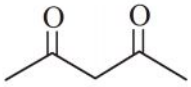
2. Ans (5)

$$2CrI_3 + 27H_2O_2 + 10NaOH \longrightarrow 2Na_2CrO_4 + 6NaIO_4 + 32H_2O$$

$$-\frac{1}{10}\frac{d[NaOH]}{dt} = -\frac{1}{2}\frac{d[CrI_3]}{dt}$$

$$\Rightarrow \frac{\frac{-d[NaOH]}{dt}}{\frac{-d[CrI_3]}{dt}} = \frac{10}{2} = \boxed{5}$$

3. Ans (1)



$$\xrightarrow[I_2]{NaOH} CHI_3 + 2CH_3-COONa$$

5. Ans (5)

i, iii, vi, vii, viii

PART-2 : CHEMISTRY

SECTION-IV

1. Ans (A->PQ,B->PST,C->PR,D->P)

Crystal system	Crystallographic axes	Crystallographic angles	Possible unit cells
Cubic (Most symmetrical)	a = b = c	$\alpha = \beta = \gamma = 90°$	3 SC, BCC, ECC
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	S, BC 2
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	S, BC, FC, EC 4
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°$ $\beta \neq 90°$ $\beta > 90°$	S, EC 2
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$ $\gamma = 120°$	S 1
Rhombohedral (Trigonal)	a = b = c	$\alpha = \beta = \gamma \neq 90°$ $90° \neq \alpha = \beta = \gamma < 120°$	S 1
Triclinic (Most unsymmetrical)	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	S 1
		Total	14

2. Ans (A->P,B->PQS,C->PR,D->QST)

(A) When r = 0, $\psi(r) \neq 0 \Rightarrow$ s-orbital

3 radial node $\Rightarrow$ P

(B) 3 radial node $\Rightarrow$ P, Q, S

(C) For s-orbitals, $\psi(\theta, \phi)$ is independent of θ and ϕ.

(D) Angular node = $\ell \Rightarrow$ Q, S

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\lim_{n\to\infty} \sum_{n=1}^{n} \left(\frac{1}{n^2 \tan^{-1} n^2} - \frac{1}{(n+1)^2 \tan^{-1}(n+1)^2} \right) = \frac{4}{\pi}$$

2. Ans (D)

$A(B + C) = KI$

$B + C = KA^{-1}$ ...(1)

$B^T + C^T = K(A^{-1})^T$

$B - C = KA$ ...(2)

$2B = K(A^{-1} + A)$

$2tr(B) = K.2\ tr(A) \Rightarrow K = 8$

3. Ans (A)

$$\frac{1}{7} \times \frac{1}{4} \cdot \left(\frac{3}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} \right) + \frac{1}{4} \cdot \frac{1}{7} \left(\frac{3}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} \right)$$

$$\frac{1}{7} \times \frac{1}{4} \times \left(\frac{9}{6} + \frac{15}{6} \right) = \frac{1}{7}$$

4. Ans (D)

$\vec{v}_1 . \vec{v}_2 = 0 \Rightarrow a = -1, 1$

$a = -1 ; \vec{v}_1 = \hat{i} - 2\hat{j}, \vec{v}_2 = 2\hat{i} + \hat{j}$

$a = 1 ; \vec{v}_1 = \hat{i} - 2\hat{j} + 2\hat{k}, \vec{v}_2 = 2\hat{i} + \hat{j}$

$\vec{v}_2$ lies in the plane $3x - \lambda y + z = 1$

$\Rightarrow \lambda = 6$

Now, volume of the parallelopiped by $\vec{v}_1$, $\vec{v}_2$ and $\vec{n}$ is

$$\left| \begin{vmatrix} 1 & -2 & 2 \\ 2 & 1 & 0 \\ 3 & -6 & 1 \end{vmatrix} \right| = |-25| = 25$$

5. Ans (A)

$x^2 - p^2 = x^2 - 4x + p^2 \Rightarrow x = \frac{p^2}{2}$

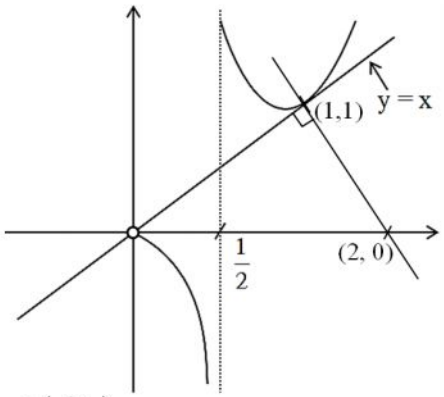
$\frac{p^4}{4} - p^2 < 0 \Rightarrow p^2(p^2 - 4) < 0 \Rightarrow p \in (-2, 2), p \neq 0$

$\therefore\ |B| = 2$

PART-3 : MATHEMATICS

SECTION-I (ii)

6. Ans (A,B,D)



$$\frac{d\left(x^3/y\right)}{1+x^3/y} = \frac{dx}{x} \Rightarrow \ell n(1+x^3/y) = \ell nx + \ell nc$$

$$1+\frac{x^3}{y} = cx$$

$$(1,1) \Rightarrow c = 2 \Rightarrow 1+\frac{x^3}{y} = 2x \Rightarrow y = \frac{x^3}{2x-1}$$

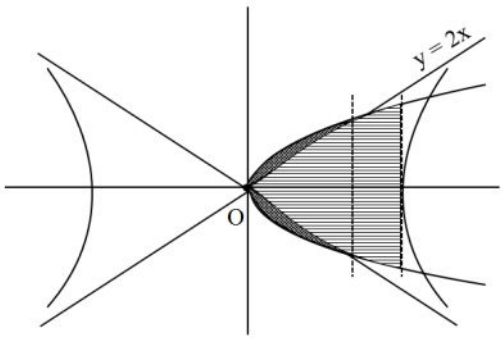
Tangent as (1,1)

$$y - 1 = 1\cdot(x-1) \Rightarrow y = x$$

Normal at (1, 1)

$$y - 1 = -(x-1) \Rightarrow x + y = 2$$

7. Ans (A,C)



For options A and B required area is $\frac{2}{3}$

For options C and D required area is

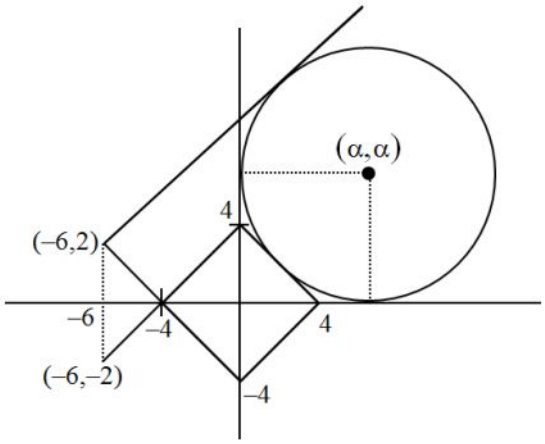
$$\left(\frac{16\sqrt{2}}{3} - \frac{2}{3}\right) \text{ sq. units}$$

8. Ans (A,D)

$$g(x) = \begin{cases} (x^2-4)^2 - 4 & x < -\sqrt{6} \text{ or } x > \sqrt{6} \\ 6 - x^2 & -\sqrt{6} \le x \le -2 \text{ or } 2 \le x \le \sqrt{6} \\ -x & -2 < x < 0 \\ x & 0 \le x < 2 \end{cases}$$

9. Ans (A)

Circle is -



$$(x-\alpha)^2 + (y-\alpha)^2 = \alpha^2$$

$$\left|\frac{\alpha+\alpha-4}{\sqrt{2}}\right| = \alpha$$

$$\Rightarrow 2\alpha - 4 = \sqrt{2}\alpha$$

$$\alpha(2-\sqrt{2}) = 4$$

$$\alpha = \frac{4(2+\sqrt{2})}{2} = 4+2\sqrt{2}$$

maximum length of the tangent will be -

$$= \sqrt{(-6-\alpha)^2 + (-2-\alpha)^2 - \alpha^2}$$

$$= \sqrt{\alpha^2 + 16\alpha + 40}$$

$$= \sqrt{(\alpha+8)^2 - 24} = \sqrt{(12+2\sqrt{2})^2 - 24}$$

$$= 4\sqrt{8+3\sqrt{2}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (140)

$$\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} \ell n|3\cos(x+\frac{\pi}{3})+5\cos x|dx$$

$$=\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} \ell n|7\sin(x+\alpha)|dx$$

$$=\pi\ell n7+\int_0^{\pi}\ell n|\sin x|dx$$

$$=\pi\ell n7+2\left(-\frac{\pi}{2}\ell n2\right)$$

$$\pi\ell n\frac{7}{2}$$

$$\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} |3\sin^2x-6\sin x\cos x+\cos^2x-2|dx$$

$$=\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} |\cos 2x+3\sin 2x|dx=2\sqrt{10}=\sqrt{40}$$

$\Rightarrow p\cdot q=140$

2. Ans (255)

$g(x^3-1)=x^5+x-2$

$\Rightarrow g(t)=(t+1)^{\frac{5}{3}}+(t+1)^{\frac{1}{3}}-2$

$g'(t)=\frac{5}{3}(t+1)^{\frac{2}{3}}+\frac{1}{3}(t+1)^{\frac{-2}{3}}$

$g(7)=32,\ g'(7)=\frac{27}{4}$

$\frac{d}{dx}((x^3+1)g(x^3))=(x^3+1)g'(x^3)3x^2+$
$g(x^3).3x^2$

$\frac{d}{dx}((x^3+1)g(x^3))|_{x=7^{1/3}}=\ell=3.7^{2/3}.86$

$\Rightarrow \ell(7^{-2/3})=258$

$\frac{d}{dx}\left(\frac{f(x)}{x^2}\right)=\frac{x^2f'(x)-f(x).2x}{x^4}$ $\quad f(-2)=-1$

$=\frac{xf'(x)-2f(x)}{x^3}$ $\quad f'(-2)=0$

$f''(-2)=0$

$\frac{d^2}{dx^2}\left(\frac{f(x)}{x^2}\right)$

$=\frac{x^3(xf''(x)+f'(x)-2f'(x))-(xf'(x)-2f(x))3x^2}{x^6}$

$f(-2)=-1,\ f'(-2)=0,\ f''(-2)=0$

$\frac{d^2}{dx^2}\left(\frac{f(x)}{x^2}\right)=m=\frac{-6}{16}=\frac{-3}{8}\Rightarrow 8m=-3$

$\therefore\ \ell(7^{-2/3})+8m=255$

3. Ans (6)

$\alpha^7=3$

$x^7-21[x^2(x+1)+x^2+x+1]+15$

$=3(1+\alpha)^7-21[(\alpha^4+2\alpha^3+\alpha^2)(\alpha^2+\alpha+1)$

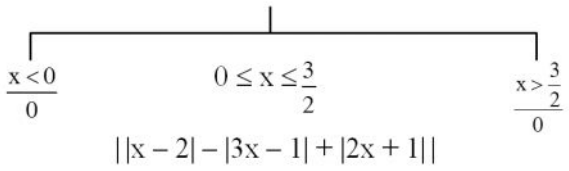
$+\alpha^4+2\alpha^3+\alpha^2+\alpha^2+\alpha+1]+15$

$=6$

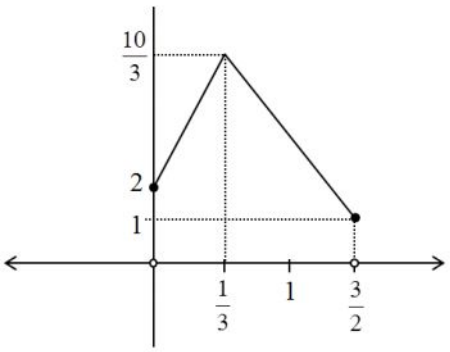
4. Ans (5)

$h(x)=||x-2|-|3x-1|+|2x+1||[|-x^2+\frac{3}{2}$

$x+1|]$



$||x-2|-|3x-1|+|2x+1||$



5. **Ans (1365)**

$x + y + z + w = 4 + 8(\sin\theta - \cos\theta)$

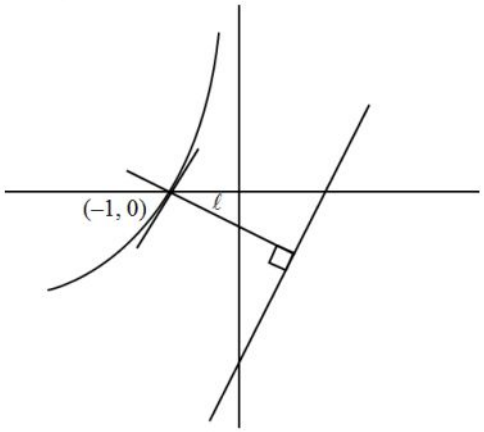
$4 \le x + y + z + w \le \underbrace{4 + 8\sqrt{2}}_{15.3}$

$\therefore$ Natural solutions corresponding to

$x + y + z + w \le 15$

$= {}^{15}C_4 = 1365$

6. **Ans (33)**



Regarding locus will be tangent to the parabola

at (1/2, –9/8) which is $y = 2x - 17/8$

$y = x - \dfrac{1}{x}$

$y' = 1 + \dfrac{1}{x^2} = 2 \Rightarrow x = \pm 1$

$\therefore \ \ell = \left|\dfrac{-2 - \frac{17}{8}}{\sqrt{5}}\right| = \dfrac{33}{8\sqrt{5}}$

$\therefore \ 8\sqrt{5}\ell = 33$

PART-3 : MATHEMATICS

SECTION-IV

1. **Ans (A->P,B->R,C->T,D->R)**

(A) $-7x + y + 15 = 0$

$x - y + 3 = 0$

$a_1a_2 + b_1b_2 = -ve$

$\therefore$ Bisector with positive sign will be acute angle bisector as well as origin containing bisector

$\therefore (-7\alpha < + \alpha^2 - 3 + 15)\,(\alpha - (\alpha^2 - 3) + 3) > 0$

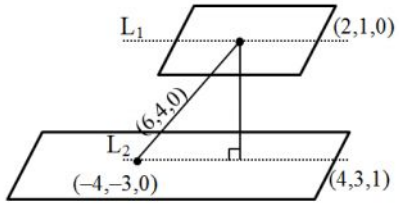
$(\alpha^2 - 7\alpha + 12)\,(\alpha^2 - \alpha - 6) < 0$

$(\alpha - 3)\,(\alpha - 4)\,(\alpha - 3)\,(\alpha + 2) < 0$

$\alpha \in (-2, 4),\ \alpha \ne 3$

(B) $n((A \times B) \cap (B \times C)) = 4$

(C)



dr's of L_1 and L_2 $\begin{vmatrix} i & j & k \\ 2 & -3 & 1 \\ 1 & -1 & -1 \end{vmatrix}$

$i(4) - j(-3) + k(1)$

$4\hat{i} + 3\hat{j} + \hat{k}$

A point on $L_1 : 2x - 3y = 1$

$3(x - y = 1)$

$-x = -2 \Rightarrow x = 2$

$y = 1$

A point on $L_2 : 2x - 3y = 1$

$3(x - y = -1)$

$-x = 4 \ \Rightarrow x = -4$

$y = -3$

Projection of $6i + 4j$ on $4\hat{i} + 3\hat{j} + \hat{k}$ is

$\dfrac{24 + 12}{\sqrt{16 + 9 + 1}} = \dfrac{36}{\sqrt{26}}$

$\therefore SD = \sqrt{52 - \dfrac{36 \times 36}{26}} = 2\sqrt{\dfrac{7}{13}}$

(D) (i) 4F $M_1M_2M_3 \longrightarrow 4!\cdot 4!$

(ii) 2F 2F $M_1M_2M_3 \longrightarrow |M_1|M_2|M_3|$

$\Rightarrow \dfrac{4!}{2!\,2!\,2!} \times {}^4C_2 \times 2!\ 2!\ 2!\ \times 3!$

$= 36(4!)$

$\therefore \ 60(4!)$

2. Ans (A->ST,B->ST,C->QRST,D->PQRST)

(A) $x^3 + (221 - 4m)x^2 + 220x + 4m$

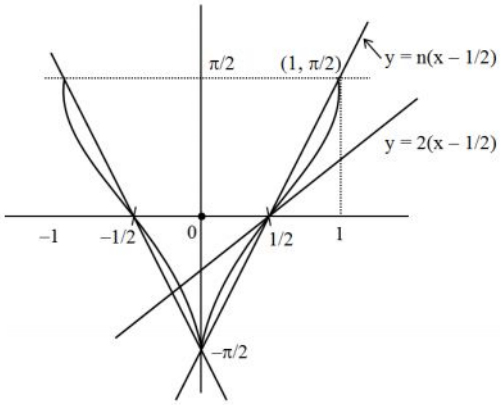
$= (x + 1)(x^2 - (4m - 220)x + 4m)$

$\widehat{\alpha\ \beta}$

$$\left.\begin{array}{l}\alpha + \beta = 4m - 220 \\ \alpha\beta = 4m\end{array}\right\} \quad \alpha = 14,\ \beta = 18$$

$$\frac{\alpha + \beta + m}{19} = \frac{14 + 18 + 63}{19} = 5$$

(B)



Clearly p must lie in $(2, \pi)$

(C)

x_i	1	2	3		n
w_i	1^2	2^2	3^2		n^2
$x_i w_i$	1^3	2^3	3^3		n^3

$$\bar{x}_w = \frac{\sum x_i w_i}{\sum w_i}$$

$$= \frac{\sum n^3}{\sum n^2} = \frac{\frac{n^2(n+1)^2}{4}}{\frac{n(n+1)(2n+1)}{6}} = \frac{3n(n+1)}{2(2n+1)}$$

$$\therefore \lim_{n\to\infty} \frac{3n(n+1)}{n \cdot 2(2n+1)} = \frac{3}{4}$$

(D) $f(x) = -x + 7 = f^{-1}(x)$

$\therefore$ $f(x) - f^{-1}(x) = 0$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,C	A,B,C,D	A,C	A,C	A,B	C,D	B,C,D
SECTION-I (ii)	Q.	8	9	10	11			
	A.	A	C	D	D			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	21.00	22.50	5.00	40.35 to 40.45	2.00	9.60 to 9.70	

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,D	B,C	A,B	B,C	C,D	B,C	B,C
SECTION-I (ii)	Q.	8	9	10	11			
	A.	D	B	A	B			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	4.00	13.66 to 13.67	9.00	5.00	-530.00	20.00	

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	B,C	A,D	A,B,C	A,B,C	A,C	A,B,D	A,B,C
SECTION-I (ii)	Q.	8	9	10	11			
	A.	A	B	B	A			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	1.75	1.51	2.82	1.80	5.50	2.00	

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C)

$$i_1 - i_2 = i_3 \; ; \; i_1 + i_2 = \frac{E}{R} \; ; \; i_2 = \frac{E}{2R} - \frac{i_3}{2}$$

$$L\frac{di_3}{dt} + i_3.2R = i_2R$$

$$L\frac{di_3}{dt} = R\left(\frac{E}{2R} - \frac{i_3}{2} - 2i_3\right) i_1$$

$$\int_0^{i_3} \frac{di_3}{\frac{E}{2R} - \frac{5}{2}i_3} = \frac{R}{L}\int_0^t dt$$

$$\frac{-2}{5}\ln\left(\frac{\frac{E}{2R} - \frac{5}{2}i_3}{\frac{E}{2R}}\right) = \frac{Rt}{2}$$

$$\frac{5}{2}i_3 = \frac{E}{2R}\left(1 - e^{-\frac{5Rt}{2L}}\right)$$

$$i_3 = \frac{E}{5R}\left(1 - e^{-\frac{5Rt}{2L}}\right)$$

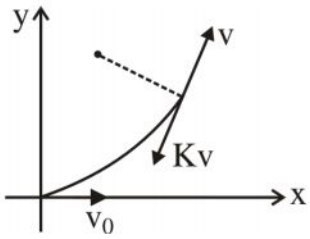
$$i_2 = \frac{E}{2R} - \frac{i_3}{2} = \frac{E}{2R} - \frac{E}{10R} + \frac{E}{10R}e^{-\frac{5Rt}{2L}}$$

$$= \frac{E}{10R}\left(4 + e^{-\frac{5Rt}{2L}}\right)$$

2. Ans (A,B,C,D)

Under normal conditions total energy, potential energy and kinetic energy in ground state and first excited state are – 13.6 eV, – 27.2 eV, 13.6 eV, – 3.4 eV, – 6.8 eV and 3.4 eV respectively. If potential energy in ground state is taken to be zero, then kinetic energy will remain unchanged but potential and total energies are increased by 27.2 eV. Therefore, the new values are 13.6 eV, 0, 13.6 eV, 23.8 eV, 20.4 eV and 3.4 eV respectively.

3. Ans (A,C)



$$r = \frac{mv}{qB},\ \omega = \frac{d\theta}{dt} = \frac{qB}{m}$$

For $\theta = 2\pi$, $\vec{v}$ is parallel to +ve x-axis.

$$\Delta t = \frac{\Delta\theta}{\omega} = \frac{2\pi m}{qB}$$

along tangential direction.

$$a_t = \frac{dv}{dt} = -\frac{Kv}{m}$$

$$\int_{v_0}^{v} \frac{dv}{v} = -\frac{K}{m}\int_0^t dt$$

$$v = v_0 e^{-\frac{k}{m}t}$$

$$\text{distance, } S = \int_0^{\Delta t} v\,dt$$

$$S = -\frac{v_0}{K/m}\left[e^{-\frac{K}{m}t}\right]_0^{\frac{2\pi m}{qB}}$$

$$S = \frac{mv_0}{K}\left[1 - e^{-\frac{2\pi K}{qB}}\right]$$

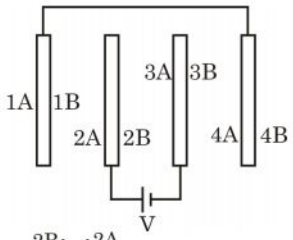
4. Ans (A,C)

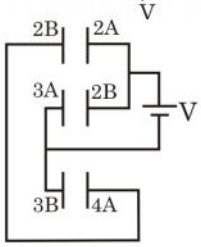
$$E_I = \frac{GMx}{R^3}\ ;\ E_{II} = \frac{GM}{R^3}\frac{R}{2}$$

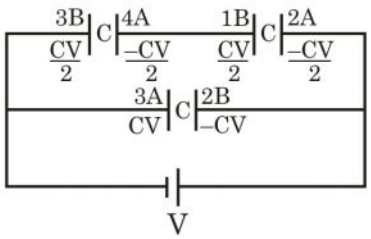
$$\frac{F_I}{F_{II}} = \frac{2x}{R}$$

$$T_I = \frac{2\pi}{4}\sqrt{\frac{R^3}{GM}}\ ;\ T_{II} = 2\sqrt{\frac{R^3}{GM}}$$

5. Ans (A,B)



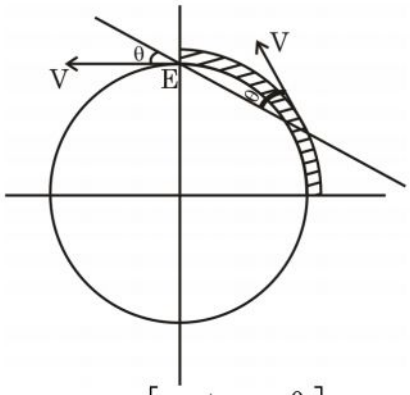




$$C_{eq} = \frac{3}{2}C = \frac{3\epsilon_0 A}{2d}$$

$$Q = C_{eq}V = \left(\frac{3\epsilon_0 A}{2d}\right)V$$

6. Ans (C,D)



$$f_{observed} = f\left[\frac{v_S + v\cos\theta}{v_s + v\cos\theta}\right] = f$$

$$\lambda_{observed} = \left(\frac{v_s + v\cos\theta}{f}\right)$$

7. **Ans (B,C,D)**

Momentum along Y-axis will remain conserved

So, $mu\cos\theta = 2m\,V_y$

$$\Rightarrow V_y = \frac{4u}{10} \qquad \ldots(i)$$

Conserving angular momentum about Y-axis during collision

$$mu\sin\theta\ell = \left(\frac{mv^2}{3} + m\ell^2\right)\omega$$

$$\Rightarrow \omega = \frac{3u\sin\theta}{4\ell} = \frac{9u}{20\ell} \qquad \ldots(ii)$$

Velocity of particle just after impact is

$$V_p = V_P\hat{j} + \ell\omega(-\hat{i}) = -\frac{9u}{20}\hat{i} + \frac{4u}{10}\hat{j}$$

for completing full rotation about Y-axis

$$W_g = \Delta k$$

$$\Rightarrow -mg(2\ell) - mg\ell = 0 - \frac{1}{2}\left(\frac{4}{3}m\ell^2\right)\omega^2$$

$$\Rightarrow \omega_{min} = \sqrt{\frac{9}{2}\frac{g}{\ell}} \Rightarrow u_{min} = \frac{20\ell}{9}\sqrt{\frac{9}{2}\frac{g}{\ell}}$$

$$10\sqrt{\frac{2}{9}g\ell} = \frac{10}{3}\sqrt{2g\ell}$$

PART-1 : PHYSICS

SECTION-I (ii)

9. **Ans (C)**

$$I = \frac{1}{2}E_0E_0^2.C$$

$$570 = \frac{1}{2}E_0 \times \frac{1}{4}\mu_0^2C^2J_{max}^2.C$$

$$= \frac{1}{8}J_{max}^2.C\mu_0$$

$$J_{max}^2 = \frac{8\times570}{3\times10\times4\pi\times10^{-7}} = \frac{2\times19}{\pi}$$

$$J_{max} = \sqrt{\frac{38}{\pi}} = 3.48$$

11. **Ans (D)**

$p_1 = 1.0$ atm, $V_1 = 1.0$L, $T_1 = 122$ K

$p_2 = 1.0$ atm, $V_2 = 2.0$L, $T_2 = 244$ K, $K = 2T_1$

$p_3 = 2.0$ atm, $V_3 = 2.0$L, $T_3 = 444$ K, $K = 4T_1$

$p_4 = 0.0157$ atm, $V_4 = 64$L, $T_4 = 122$ K, $K = T_1$

For $1 \longrightarrow 2$,

$$Q_1 = nC_p\Delta T = \frac{7}{2}nR(T_2 - T_1)$$

$$= \frac{7}{2}\times n\times R(T_1)$$

For $2 \longrightarrow 3$,

$$Q_2 = nC_V\Delta T = \frac{5}{2}nR(T_3 - T_2)$$

$$= \frac{5}{2}\times n\times R(2T_1) = 5nRT_1$$

For $3 \longrightarrow 4$,

$Q_3 = 0$

For $4 \longrightarrow 1$,

$$Q_4 = nRT\ell n\frac{V_2}{V_1}$$

$$= nRT_1\ell n\frac{1}{64} = -nRT_1\ell n64$$

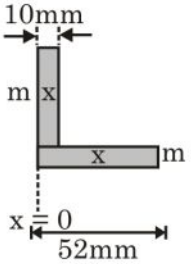
$$\eta = 1 - \frac{Q_-}{Q_+} = 1 - \frac{nRT_1\ell n64}{\frac{7}{2}nRT_1 + 5nRT_1}$$

$$= 1 - \frac{6\ell n2}{\frac{17}{2}} = 1 - \frac{12}{17}\ell n2$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (21.00)**



COM of the two dominoes is $x_0 = \frac{m(5) + m(26)}{2m}$

$x_0 = 15.5$ mm

For system to remain in equilibrium the COM of upper two dominoes must be over the thickness of third domino.

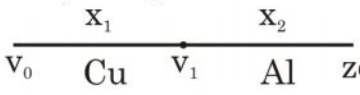
$x < x_0 < x + 10$

or $x < 15.5$ mm

and $15.5 < x + 10 \Rightarrow x > 5.5$ mm



2. Ans (22.50)



The resistive force is proportional to velocity

$ma = -kv$

$\Rightarrow \frac{mdv}{dt} + kv = 0$

$\Rightarrow \frac{d}{dt}(mv + kx) = 0$

$\Rightarrow \frac{d}{dt}\left(\frac{m}{k}v + x\right) = 0$

$\Rightarrow \frac{m}{k}v + x = C$ (constant) .....(1)

If initial velocity is v_0 and maximum stopping distance is x_{max}

$C = \frac{mv_0}{k} = x_{max}$ .....(2)

$\Rightarrow x_{max} = \mu v_0$ .....(3)

where $\mu = \frac{m}{k}$ is the characteristic of the metal plate causing the breaking.

Hence $\mu_{Al}v_0 = 30$ cm and $\mu_{Cu}v_0 = 20$ cm

$\frac{\mu_{Al}}{\mu_{Cu}} = \frac{3}{2}$

Now, $\mu v + x = C = \mu v_0$

$\Rightarrow \mu v_1 + 15 = \mu v_0$

$v_1 = v_0 - \frac{15}{\mu_{Cu}}$

and $x_2 = \mu_{Al}v_1 = \mu_{Al}\left(v_0 - \frac{15}{\mu_{Cu}}\right)$

and $x_2 = \mu_{Al}v_1 = \mu_{Al}\left(v_0 - \frac{15}{\mu_{Cu}}\right)$

$\Rightarrow x_2 = \mu_{Al}v_0 - \frac{\mu_{Al}}{\mu_{Cu}} \times 15$

$= 30 - \frac{3}{2} \times 15 = 7.5$ cm

$x_1 + x_2 = 15 + 7.5 = 22.5$ cm

3. Ans (5.00)

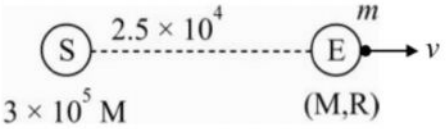
$\Delta = 70\alpha$

$2d \sin\alpha + 2d \sin\Psi = 70\alpha$

$10 \times 10^{-3} \times \frac{10^{-3}}{1} + 10 \times 10^{-3} \cdot \frac{y}{2} = 70\alpha$

$y = 5$ mm

4. Ans (40.35 to 40.45)



For earth, $V_e = \sqrt{\frac{2GM}{R}}$

For Sun + Earth,

Potential energy $= \left(-\frac{GM}{R} - \frac{3 \times 10^5}{2.5 \times 10^4}\frac{GM}{R}\right)m$

$= \frac{GM}{R}(1 + 12) = \frac{13GMm}{R}$

$\therefore \frac{1}{2}mv^2 = \frac{13GMm}{R}$

$v = \sqrt{\frac{2GM}{R}} \cdot \sqrt{13} = \sqrt{13}v_e$

$\therefore v_e = 40.4$ km/s

5. Ans (2.00)

$\frac{dP}{dy} = -\rho g$

$dP = -\rho g dy$

$\left[P = \frac{M}{VM_0}RT = \frac{\rho RT}{M_0} \Rightarrow \rho = \frac{PM_0}{RT}\right]$

$dP = -\frac{PM_0}{RT}dy$

$dP = -\frac{PM_0}{R}\frac{dy}{(T_0 - \alpha y)}$

$\int_{P_0}^{P}\frac{dP}{P} = -\frac{M_0}{R}\int_0^y\left(\frac{dy}{T_0 - \alpha y}\right)$

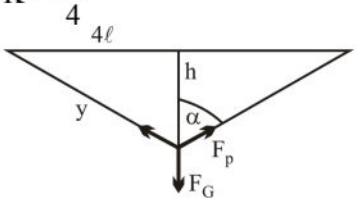
$\Rightarrow P = P_0\left(1 - \frac{\alpha y}{T_0}\right)^{\frac{M_0 g}{\alpha R}}$

6. Ans (9.60 to 9.70)

Since the transverse fibers do not affect the bending of the web, we can only deal with the elongation of the radial fibers.

If a part of the fiber has a length ℓ and a stiffness k, then it is extended by $\Delta x = F/k$ by a force of magnitude F. The radial fiber consists of 4 such parts, therefore it is lengthened by the force $F\Delta X = 4\Delta x$. The total stiffness of the radial fiber will then be (similarly to four springs suspended in series

$$K = \frac{k}{4}$$



In the middle of the web at point O, the spider acts on the web with its weight, which is in balance with the six elastic forces Fp

$mg = 6F_p\cos\alpha,$

$$F_p = K(y - 4\ell) = \frac{k}{4}\left(\sqrt{(4\ell)^2 + h^2} - 4\ell\right)$$

$$\cos\alpha = \frac{h}{y} = \frac{h}{\sqrt{(4\ell)^2 + h^2}},$$

$$mg = 6\frac{k}{4}\left(\sqrt{(4\ell)^2 + h^2} - 4\ell\right)\frac{h}{\sqrt{(4\ell)^2 + h^2}}$$

$$k = \frac{2mg\sqrt{(4\ell)^2 + h^2}}{3h\left(\sqrt{(4\ell)^2 + h^2} - 4\ell\right)}$$

$$k = \frac{2mg}{3h} \times \frac{4\ell}{4\ell\left(\frac{\sqrt{1+h^2}}{(4\ell)^2} - 1\right)}$$

$$= \frac{2mg}{3h}\frac{1}{\left(1 + \frac{1}{2}\frac{h^2}{(4\ell)^2} - 1\right)} = \frac{2mg}{3h} \times \frac{2 \times 16\ell^2}{h^2}$$

$$= \frac{64}{3}\frac{mg\ell}{h^3}$$

$$= \frac{64}{3} \times \frac{0.01 \times 10 \times 0.1 \times 0.1}{125 \times 10^{-9}} = 0.17 \times 10^6 \text{N/m}$$

The modulus of elasticity in tension will then be according to Hooke's law

$$E = \frac{\sigma}{\varepsilon} = \frac{F_p}{S\frac{\Delta x}{\ell}} = \frac{4k\ell\Delta x}{\pi d^2\Delta x} = \frac{4k\ell}{\pi d^2}$$

$$E = \frac{4 \times 0.17 \times 10^6 \times 0.1}{\frac{22}{7} \times (0.15 \times 0.15) \times 10^{-12}}$$

$$= \frac{0.68 \times 10^5}{0.07065 \times 10^{-12}} = 9.62 \times 10^{17} \text{ Pa}$$

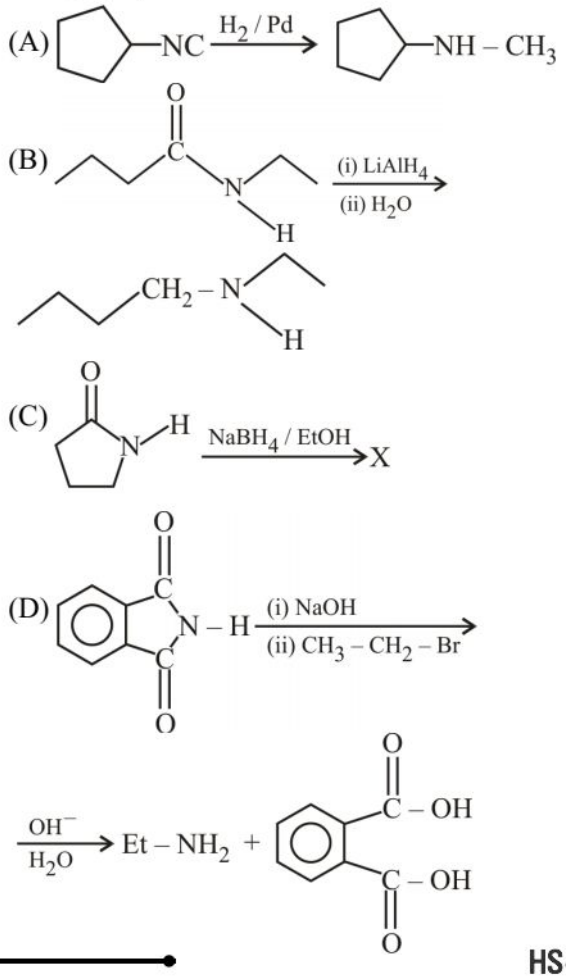
PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B,C)

Coordination entity	Wavelength of light absorbed (nm)	
$[CoCl(NH_3)_5]^{2+}$	(P)	535
$[Co(NH_3)_5(H_2O)]^{3+}$	(Q)	500
$[Co(NH_3)_6]^{3+}$	(R)	475
$[Co(CN)_6]^{3-}$	(S)	310
$[Cu(H_2O)_4]^{2+}$	(T)	600

3. Ans (A,B)



6. Ans (B,C)

$P_A = 235\ y_A - 125\ x_B\ y_A$

$= y_A\ [235 - 125\ (1 - x_A)]$

$= y_A\ [235 - 125 + 125\ x_A]$

$P_A = y_A\ [110 + 125\ x_A]$

$P_A = P_T \times y_A$

$\Rightarrow P_T = (P^\circ_A - P^\circ_B)\ x_A + P^\circ_B$

$\therefore P^\circ_B = 110$

$P^\circ_A - P^\circ_B = 125$

$P^\circ_A = 235.$

7. Ans (B,C)

(A) Coagulation value.

(B) Coagulation power $\propto \dfrac{1}{\text{Coagulation value}}$

(C) Hardy Schultz rule

(D) Definition of protective colloid.

PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (D)

$\Delta G = -nFE_{cell} = \dfrac{-2 \times 96500 \times 0.059}{1000}$ KJ mol^{-1}.

9. Ans (B)

$0.059 = \dfrac{-0.059}{2} \log \dfrac{[M^{2+}]_A}{0.001}$

or, $\dfrac{[M^{2+}]_A}{0.001} = \dfrac{1}{100}$ or, $[M^{2+}]_A = 10^{-5} = s$(say)

$MX_2(s) \rightleftharpoons M^{2+}(aq) + 2X^-(aq)$

s 2s

$K_{sp} = 4s^3 = 4 \times 10^{-15}$

10. Ans (A)

$n = 2 \longrightarrow \ell =$ 0 1 2 3

s p d f

No change in the volue of m

$\therefore$ total orbitals associated to 2nd shell = 16

The value of $s = +\frac{1}{4}, +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{4}$

$\therefore$ total no of e^- associated to 2nd shell $= 16 \times 4 = 64$

11. Ans (B)

p-orbital having $4e^-$ (according to value of 's')

PART-2 : CHEMISTRY

SECTION-II

2. Ans (13.66 to 13.67)

(i) $Ag_2SO_3 + H^+ \longrightarrow 2Ag^+ + SO_2 + H_2O$

(ii) $FeSO_4$ solution absorbs NO and becomes brown because of the formation of $[Fe(NO)]^{2+}SO_4^{2-}$.

(iii) Being covalent do not dissociate

(iv) Ammonium hydroxide in presence of ammonium chloride is group reagent of III[rd] group.

(v) $2[Cr(OH)_4]^-$ (green colouration) + $3H_2O_2$ + $2OH^-$

$\longrightarrow 2CrO_4^{2-} + 8H_2O$

(vi) Red colour solution or precipitate is produced when reagent reacts in alkaline solution.

(vii) $HO.SO_2.NH_2 + HNO_2$

$\longrightarrow N_2 + 2H^+ + SO_4^{2-} + H_2O$

(viii) $Fe^{2+} + [Fe(CN)_6]^{3-}$

$\longrightarrow Fe^{3+} + [Fe(CN)_6]^{4-}$

$4Fe^{3+} + [Fe(CN)_4]^{4-}$

$\longrightarrow Fe_4[Fe(CN)_6]_3 \downarrow$(Turnbull's blue)

(ix) $4Cr(OH)_3\downarrow + 5O_2^{2-}$

$\longrightarrow 4CrO_4^{2-}$ (yellow solution) + $6H_2O$

a = 5 b = 4 $\dfrac{a^2 + b^2}{3} = \dfrac{25 + 16}{3} = 13.67$

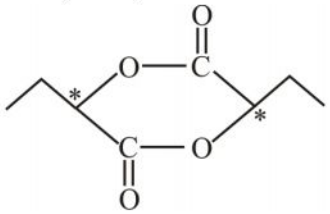
3. Ans (9.00)

x = 5, y = 4

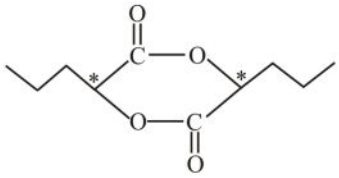
Tollen's test : (i), (ii), (iv), (vi), (vii)

Fehling test : (i), (ii), (iv), (vi)

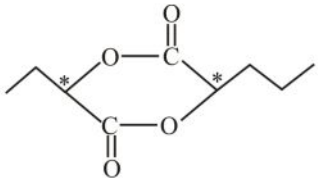
4. **Ans (5.00)**



(3)



(3)



(4)

5. **Ans (-530.00)**

$dG = V.dP - S.dT$; at const. volume

$\Delta G = V.\Delta P - (10 + 10^{-2}\,T).dT \quad \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow$

$P_2 = 1 \times 400/300$

$\Delta G = 24.6 \times \left(\frac{4}{3} - 1\right) \times 100 - \left[10 \times 100 + 10^{-2} \times \left(\frac{T_2^2}{2} - \frac{T_1^2}{2}\right)\right]$

$\Delta G = 24.6 \times \frac{1}{3} \times 100 - [1000 + 10^{-2} \times \left(\frac{160000}{2} - \frac{90000}{2}\right)]$

$= 24.6 \times \frac{1}{3} \times 100 - 1000 - 350 = -530$ J

6. **Ans (20.00)**

$AB \underset{}{\overset{2\times10^{-10}}{\rightleftharpoons}} A^+ + B^-$

$s \quad \approx 0$

$B^- + H^+ \xrightarrow{10^8} HB$

$\approx 0 \quad 10^{-3} \qquad s$

$2 \times 10^{-10} = s \times [B^-]$ ..... (1)

$10^8 = \frac{s}{[B^-] \times 10^{-3}} \Rightarrow 10^8 = \frac{s^2}{2 \times 10^{-10} \times 10^{-3}}$

$\Rightarrow s^2 = 2 \times 10^{-5} \Rightarrow s = \sqrt{20} \times 10^{-3}$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C)**

$\text{Required Area} = 2\int_{-3/2}^{0} 2\left(x + \frac{3}{2}\right) dx = 2\sqrt{3}$

2. **Ans (A,D)**

$A = \int_0^1 [-\ell ny) - (e^y - e]\, dy = 2$

3. **Ans (A,B,C)**

$3x^2 \frac{dx}{dy} = \frac{x^3}{x^3 - y} \qquad x^3 = t$

$\frac{dt}{dy} = \frac{t}{t - y} \Rightarrow tdt = ydt + xdy$

$\frac{t^2}{2} = t.\,y + \frac{c}{2}$

$\Rightarrow x^6 = 2x^3y + c$

4. **Ans (A,B,C)**

$\frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$

$= \frac{1}{2^n} + \frac{3n^2 + 10n + 6}{2^n(n^2 + 2n + 2)(n^2 - 2n + 2)}$

$= \frac{1}{2^n} + \frac{4n^2 + 8n + 8 - n^2 + 2n - 2}{2^n(n^2 + 2n + 2)(n^2 - 2n + 2)}$

$= \frac{1}{2^n} + \frac{1}{2^{n-2}(n^2 - 2n + 2)} - \frac{1}{2^{n-2}(n^2 + 2n + 2)}$

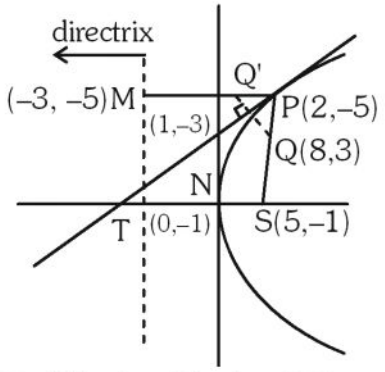
$\therefore \sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n.(n^4 + 4)}$

$= \sum_{n=2}^{\infty} \frac{1}{2^n} + \sum_{n=2}^{\infty} \left(\frac{1}{2^{n-2}(n^2 - 2n + 2)} - \frac{1}{2^n(n^2 + 2n + 2)}\right)$

$= \frac{1}{2} + \frac{6}{10} = \frac{11}{10}$

5. Ans (A,C)

Equation of tangent at P is $2x + y + 1 = 0$



N will be the mid point of PT

$\Rightarrow$ T is (0, –1)

Image of Q(8, 3) w.r.t. to tangent at P lies on PM.

Image Q' of Q(8, 3) w.r.t. $2x + y + 1 = 0$

$$\frac{x-8}{2} = \frac{y-3}{1} = \frac{-2[16+3+1]}{5}$$

$\Rightarrow$ $x = -8, y = -5$

Q'(–8, –5)

$\Rightarrow$ PM will be $y = -5$

$\Rightarrow$ axis of parabola will be $y = -1$

PQ is a focal chord of parabola equation of PQ is $4x - 3y - 23 = 0$ focus will be point of intersection axis & PS

$\Rightarrow$ focus S is (5, –1) Point M lies on directrix of parabola N will be mid point of M & S

$\Rightarrow$ M is (–3, –5)

$\Rightarrow$ directrix is $x = -3$

Normal at other end of focal chord at P will be parallel to tangent at P.

Circle described on focal radii as diameter touches tangent at vertex of parabola.

6. Ans (A,B,D)

Let $(x^2 - 4x + 6) = t$

$f(t) = t^2 - (k+2)t + k + 5 = 0$

Apply conditions of location of roots on equation $f(t) = 0$.

for exactly 4 real solution $k \in (4, 5)$

for exactly 3 real solutions $k = 5$

for exactly are real solution $k \in \phi$

for no real solutions $k \in (-\infty, 4)$

7. Ans (A,B,C)

$$\lim_{x\to 0} \frac{a\left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \ldots\right) + b\left(x - \frac{x^3}{3} + \frac{x^5}{5} + \ldots\right) + c\left(x + \frac{x^3}{6} + \frac{9x^5}{5!} + \ldots\right)}{x^5} = 3$$

$$\lim_{x\to 0} \frac{(a+b+c)x + \left(\frac{a}{3} - \frac{b}{3} + \frac{c}{6}\right)x^3 + \left(\frac{2a}{15} + \frac{b}{5} + \frac{9c}{5!}\right)x^5 + \ldots}{x^5} = 3$$

$\Rightarrow$

$a + b + c =$ ...(1)

$2a - 2b + c = 0$ ...(2)

$$\frac{2a}{15} + \frac{b}{5} + \frac{3c}{40} = 3$$ ...(3)

$\Rightarrow$ $a = 30, b = 10, c = -40$

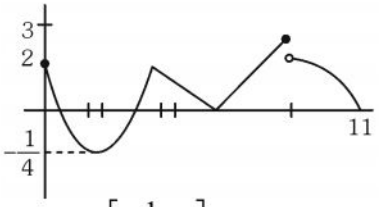
$2a + b + c = 60 + 10 - 40 = 30$

PART-3 : MATHEMATICS

SECTION-I (ii)

8. Ans (A)

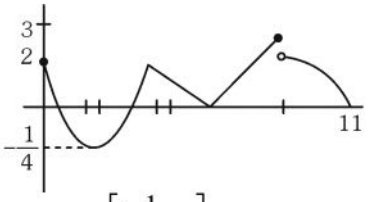
Graph of f(x)



Range : $\left[\frac{-1}{4}, 3\right]$

9. Ans (B)

Graph of f(x)



Range : $\left[\frac{-1}{4}, 3\right]$

10. **Ans (B)**

(A) $\frac{d}{dx}f(x) = \frac{e^{\sin x}}{x}$

$f(x) = \int \frac{e^{\sin x}}{x}\,dx$

$\Rightarrow \int_1^4 \frac{2e^{\sin x^2}}{x}\,.\,dx = \int_1^4 \frac{e^{\sin(x^2)}}{x^2}\,d(x^2)$

$x^2 = t$

$= [f(t)]_1^{16}$

$= f(16) - f(1)$

$\Rightarrow \boxed{k = 16}$

(B) $xf'(x) = e^{\sin x}$

$xf''(x) + f'(x) = e^{\sin x}\,.\,\sin x$

$= g(x)$

11. **Ans (A)**

(A) $\frac{d}{dx}f(x) = \frac{e^{\sin x}}{x}$

$f(x) = \int \frac{e^{\sin x}}{x}\,dx$

$\Rightarrow \int_1^4 \frac{2e^{\sin x^2}}{x}\,.\,dx = \int_1^4 \frac{e^{\sin(x^2)}}{x^2}\,d(x^2)$

$x^2 = t$

$= [f(t)]_1^{16}$

$= f(16) - f(1)$

$\Rightarrow \boxed{k = 16}$

(B) $xf'(x) = e^{\sin x}$

$xf''(x) + f'(x) = e^{\sin x}\,.\,\sin x$

$= g(x)$

(C) $I = \int_0^{\pi/2} x\cos x\,.\,\frac{e^{\sin x}}{x}\,dx$

$\sin x = t$

$\cos x\,dx = dt$

$= \int_0^1 e^t\,dt = e^1 - 1$

$\lambda = \frac{\pi}{2}$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.75)**

If the one of the digit first or last is zero then number of numbers is $8 \times 7 = 56$.

If none of the first and last digit is zero then number of numbers are $2 \times 8 \times 7 \times 7$

Hence total numbers $= 8 \times 7 + 2 \times 8 \times 7 \times 7 = 840$

2. **Ans (1.51)**

$a = -\sqrt{2},\ b = \sqrt{2}$

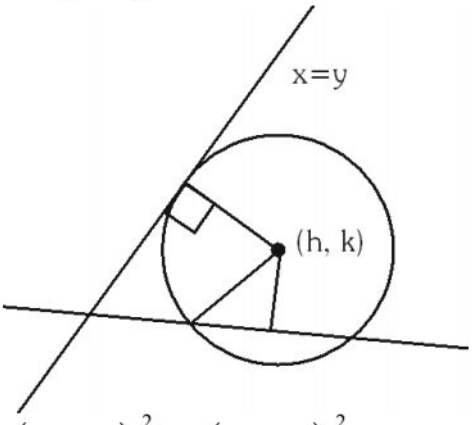
$\text{area} = -\int_{-\sqrt{2}}^{\sqrt{2}} (x^4 - 2x^2)\,dx$

$= -2\int_0^{\sqrt{2}} (x^4 - 2x^2)\,dx$

$= 2\int_0^{\sqrt{2}} (2x^2 - x^4)\,dx$

$\therefore \text{Area} = \frac{16\sqrt{2}}{15} = 1.51$

3. **Ans (2.82)**



$\left(\frac{h-k}{\sqrt{2}}\right)^2 = \left(\frac{h+k}{\sqrt{2}}\right)^2 + 2$

$\therefore\ xy = -1$

Hence locus is rectangular hyperbola with length of latus rectum $= 2\sqrt{2}$

4. Ans (1.80)

$$\lim_{n\to\infty} n\sum_{r=1}^{n} \frac{1}{3n^2 + 8rn + 4r^2}$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}\left(\frac{1}{3 + 8\left(\frac{r}{n}\right) + 4\left(\frac{r}{n}\right)^2}\right)$$

$$= \frac{1}{4}\int_0^1 \frac{dx}{x^2 + 2x + 1 - \frac{1}{4}} = \frac{1}{4}\int_0^1 \frac{dx}{(x+1)^2 - \left(\frac{1}{2}\right)^2}$$

Thus $\frac{1}{4}\ell n\left|\frac{x + 1 - \frac{1}{2}}{x + 1 + \frac{1}{2}}\right| = \frac{1}{4}\ell n\frac{9}{5}$

5. Ans (5.50)

$$\sigma_{12}^2 = \frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1d_1^2 + N_2d_2^2}{N_1 + N_2}$$

Here σ_{12}^2 = combined variance

$N_1 = N_2 = 5$, $\sigma_1^2 = 4$, $\sigma_2^2 = 5$, $d_1 = -1$, $d_2 = 1$

6. Ans (2.00)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix}$$

$\therefore\ a^2 + b^2 = 1, c^2 + d^2 = 1, ac + bd = 0$

Let $a = \cos\theta$, $b = \sin\theta$, $c = \cos\phi$, $d = \sin\phi$

$\therefore\ \cos\theta\cos\phi + \sin\theta\sin\phi = 0,\ \therefore\ \theta - \phi = \frac{\pi}{2}$

$\Rightarrow\ |a| + |b| + |c| + |d| = 2\,(|\cos\theta| + |\sin\theta|$

since $|\cos\theta| + |\sin\theta| \le \sqrt{2}$

greatest value of $2\,(|a| + |b|) = 2\sqrt{2}$